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Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

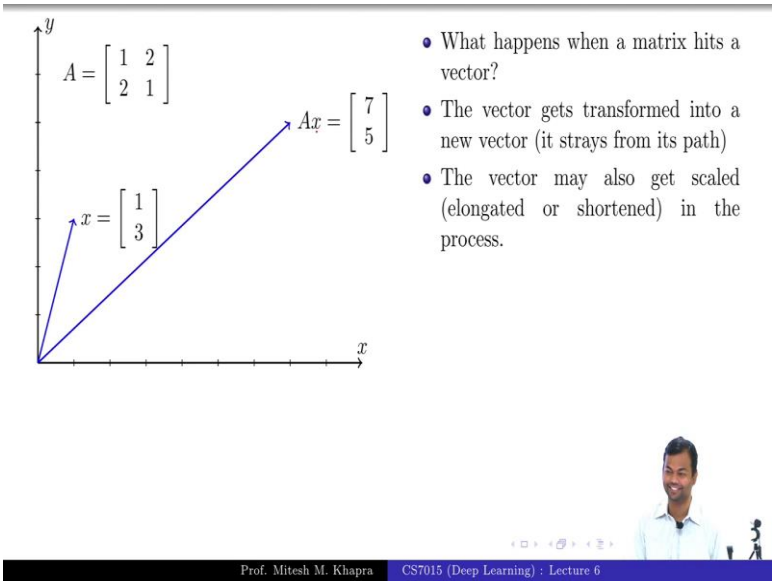
Lecture - 06
Eigen Values, Eigen Vectors, Eigen Value Decomposition, Principal Component Analysis, Singular Value Decomposition

So, this lecture actually is a bit of a digression, and it is supposed to cover some of the basics that we need for various sections of the course. So, it is very important that you understand some concepts for linear algebra specifically eigenvalues, eigenvectors and in particular. Today we will do principal component analysis, and the reason that I do it is there is an very neat relation of PCA and to autoencoders. An autoencoder is something that well cover in the course, it is a part of any deep neural network course.

And singular value decomposition is something that we using when we learn word vectors. The word vector is again something very important, I can just I can do the non SVD version of it where I just talk about what word to wick is, but that will not give you the same probably not the same interpretation as if you start from SVD and then reach word vectors, right? So, that is why I am covering these basics.

So, how many of you know eigenvalues and eigenvectors? Very embarrassing question how many of you absolutely hate eigenvalues and eigenvectors. So, let us see if we can change that today, I mean on the positive side.

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So, what happens when a matrix hits a vector? So, most of you a lot of people that I talk to right actually think that eigenvectors are the villains of linear algebra, it is very hard to understand them and so on. But today I am going to make a case for they are not the villains they are actually the superheroes of linear algebra. So, that is what the lecture is about. So, what happens when a matrix hits a vector?

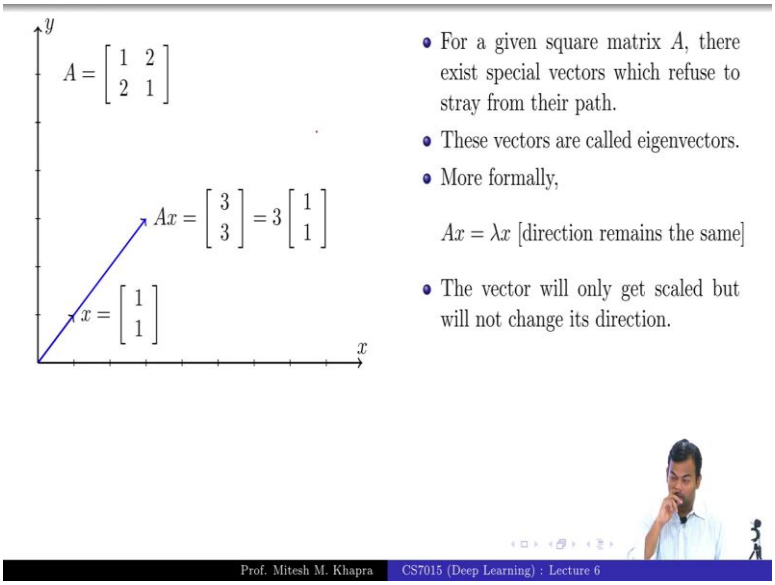
Student: Transforms it.

Transforms it right; so, actually what happens is that it strays from it is path. So, this is the original (Refer Time: 01:58) this is the original vector x ok and now once I multiply it by A; that means, if I do the transformation A x then I get a new vector. And two things happen right, one is the direction changes which is obvious, and in many cases the scale also changes; that means, the vector might get elongated it is magnitude would increase or it would decrease.

So, if you really think about it actually right? So, matrices are the real villains of linear algebra right, and we just look at this vector was minding it is own business going along it is own direction a metric comes and hits it and completely changes it is world right, I mean; it just throws it off path increases a dimension or slows it down or whatever it. So, that is they are the bad guys now for every villain what do you have a super hero right? So, what is a super hero corresponding to orbit? What does a super hero do? Know that

is a very linear algebra. I am talking about comic books that this is very linear algebraic answer he stands up to the villain right.

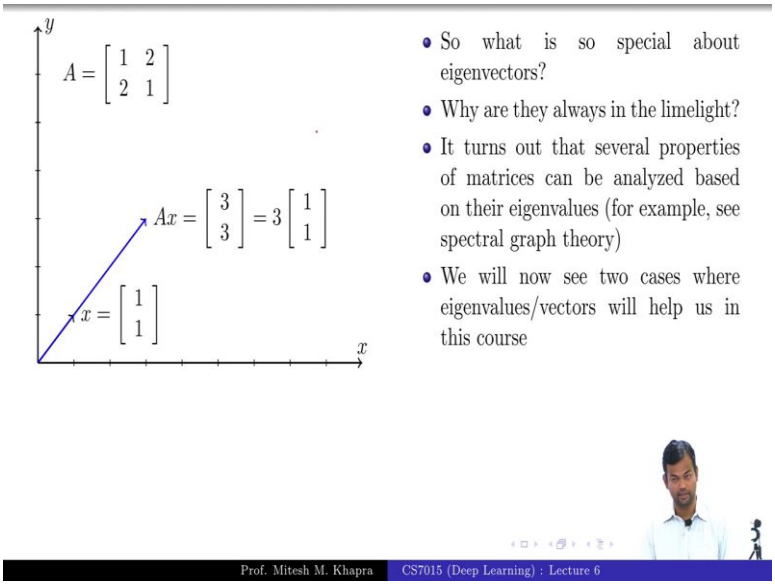
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And that is exactly what Eigen vectors do it right, they refused to change their part they tell the matrix. You can hit me as many times as you want probably you can increase my you could probably slow me down a bit or push me ahead or something, but I am not going to stray off from your path right? So, that is what eigenvalue eigenvectors do.

So here is a matrix, which is a villain and here is an eigenvector which is our hero and now when this matrix hits this eigenvector it refuses to stray from it is part right? It says I will move forward I will move back whatever, but I will not change my direction ok. I will just stay honest to what I am and these vectors are called the eigenvectors. I am more formally you can write it as Ax is equal to lambda x right so; that means, the direction remains the same only the scale changes it will either get slowed down or it will get boosted up right? So, the magnitude would change, but the direction remains the same.

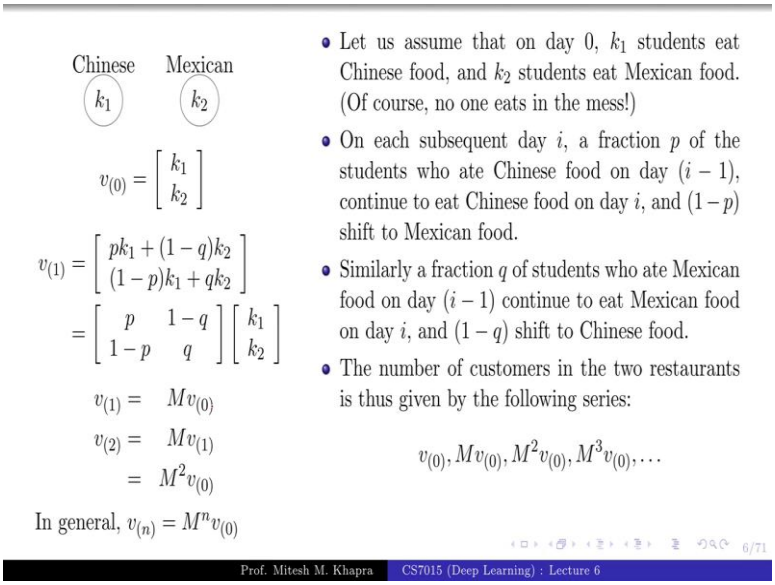
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Now, what is so special about eigenvectors? Like why are why is it that, they are always in the lime light? I know the any course that you do invariably touch eigenvectors or eigenvalues at some point in that course right, where be it machine learning, image processing whatever you do you always speech everything that you do, you will always have eigenvectors and eigenvalues, why is it so, well it is turns out that several properties of matrices can actually be explained away by looking at their eigenvalues. So, if I look at a matrix I would probably not be able to comment much on it, but if you tell me something about the eigenvalues.

I can see a lot of things about of it and there is an entire field on this way this entire spectral graph theory which looks at properties of Laplacian matrices and come in something on the properties of the graph and so on right? And that is just an example which we do not care about, but what we care about in this course there are a few things that we care about with respect to eigenvalues and eigenvector. And that is what I am going to focus on right? So, that is what this lecture is going to be out. And I will take 2 specific cases which are very important for us to understand certain concepts later on. So, I will start with the first one.

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And I will start with a very simple example to motivate this problem. And eventually will lead to a result which will help us understand a very important concept in deep neural network training which is exploding and vanishing, vanishing gradient. We will not touch that concept today, but we will use these ideas when we are looking at that later on.

So, let us take this example of 2 restaurants. So, there is a Chinese restaurant and a Mexican restaurant. And on day one k 1 students eat in the Chinese restaurant and k 2 students eat in the Mexican restaurant. So, this is what my situation is on day 0, k 1 for Chinese and k 2 for Mexican. Now what happens as is obvious people get bored or they have different want to try out different things. So, on day two or other each subsequent day what happens is that, a fraction p of the students who ate Chinese today will opt for max Mexican, on day on the next day and a fraction q of the students who ate ma Mexican today are going to opt for Chinese.

So, you get this situation right? So, I started with k 1, k 2. So, what I am saying is on day one that is the next day only a fraction p of the k 1 students will remain for Chinese and a fraction 1 minus q would be transferred from Mexican to Chinese ok. And similarly only a fraction q of the students would again stick to the Mexican food and a fraction 1 minus p into k 1 would shift from Chinese to Mexican is this setup clear ok. Can you write this

as a matrix operation it would be a matrix multiplied by a vector right can you tell me the vector.

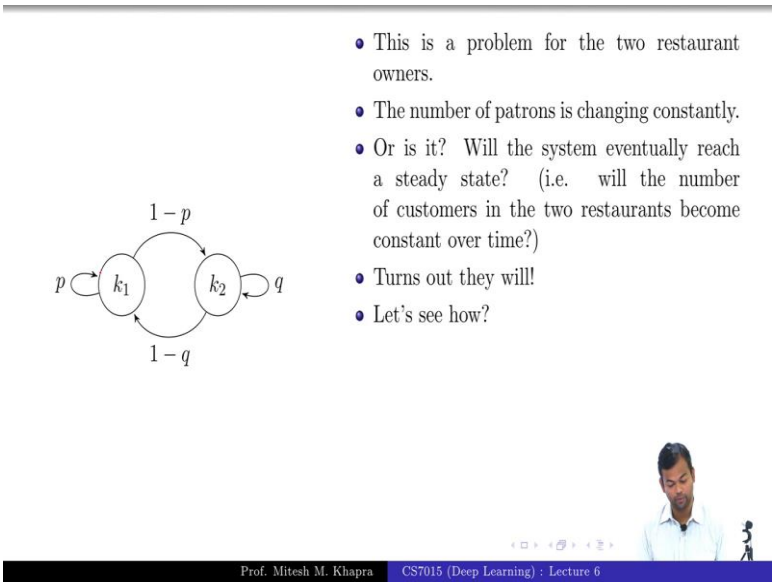
Student: (Refer Time: 06:29).

k 1 k 2 k 1 k 2 and the matrix is in all this ok, this is what it is. And I am saying that this happens on each subsequent day, it is every day now this keeps happening. So, on day 1 I started with say 180 and now day 2 it change to something again day 3 it will change something by the same fraction.

Now, let me call this as matrix M and this is of course, v 0 right by definition as we decided now what would happen on day 2 what would v 2 be M applied to v 1 right and which would be M square applied to v 0. I am just substituting the value of v 1 which is M into v 0 in general on the nth day what would happen M raised to n into v 0 ok. So, you see that the number of customers in the 2 restaurants is given by this series you had v 0 then M into v 0 then M square v 0 and so on up to M raised to n vn ok. You see how the number of customer is changing.

Now, and this is how I represent it as a state transition diagram right? So, I had certain numbers on day 1 and it changed with the trans with the probability p they will stay back with a probability 1 minus p they will move to the next or the different restaurant and so on right.

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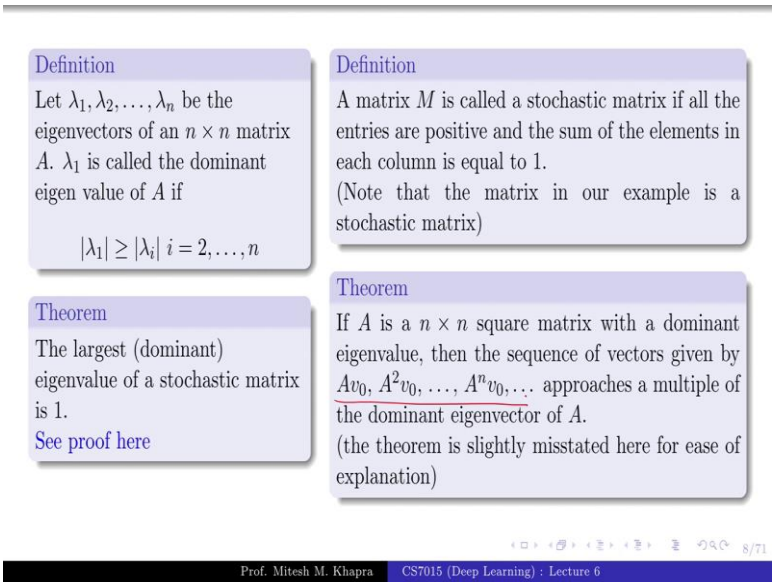


And now this though a very toyish example can you relate it to many things in real life or many things that you will take in decision making right that you are. So, even if you are playing a game for example, and even if you are playing Atari games or something, you are in a certain state based on some action that will take will move to a different state and so on right? So, these things happen in various real world applications right there is a certain state for example, even in stock market prediction, you are at a certain value of fish stock it might change to a different value right and these values you could just say them as high low or neutral that I am not going into the actual numbers.

Today the stock value is high it does it possibility that it will transition to something low and so on right? So, these kind of state transition diagrams occur in various real world examples. Now this is a problem for the two restaurant owners right, why is this a problem for the two restaurant owners? They do not know how much food to make, but every day the number of customers is changing right, but is the number of customers actually changing. Will the system eventually reach a steady state? Will it is it obvious that it will reach a steady state or maybe it will not even reaches steady, but the way I describe it I do not see why it should reach a steady state right you have some people here they go there come back go there and so on.

The only thing which I have assumed is that the transition matrix which was the matrix M is constant across all the time steps right? So, every day it is at the same priorities by which things are changed right? So, what is your guess if I were to ask you to take a guess ok. Let us see how many of you think and it is there is no correct answer here at this point. So, just tell me how many of you think it will reach a steady state? How many of you think it will keep changing and why is the sum never equal to 1 ok. So, fine so it turns out that they will right and let us see how.

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So, we will define some things and some of these are just definitions some of them have accompanying proofs, which I am not going to do here you can the proofs have been linked from the slides. So, you can take a look at them if you are interested.

So, suppose there is a matrix A n cross n matrix which has eigenvalues are lambda one lambda 2 up to lambda n. Now what this definition is saying is that, assume that there is one eigenvalue which is greater there is no assumption actually the eigenvalue which is greater than all the other eigenvalues is called the dominant eigenvalue. And when I am looking at a dominant eigenvalue I am only concerned with the magnitude not the sign. So, it could be that an eigenvalue is minus 10 and all the other eigenvalues are 1 2 3 4 5. So, the dominant eigenvalue would be minus 10 right and I will just take it as step is that clear the definition of a dominant eigenvalue.

Now, how many of you know what is the stochastic matrix? So, matrix M is called a stochastic matrix, if all the entries are positive and the sum of the elements in each column is equal to 1. So now, this definition is again slightly misstated. So, there is a row stochastic matrix the column stochastic matrix and also doubly stochastic matrix right? So, what I am talking about here is a column stochastic matrix like our matrix have you seen such a stochastic matrix any time in your life in the last 5 minutes the M matrix right? So, the M matrix is a stochastic matrix because the sum of the columns was 1

right, you had p 1 minus p q 1 minus q ok or was it some of the rows was 1 rows was 1 is it the columns.

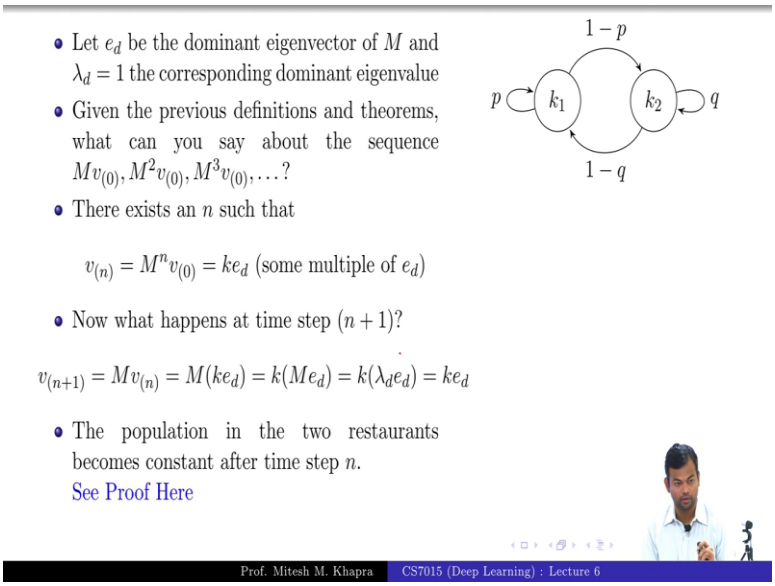
So, this is a stochastic matrix just a definition. Now I combine these two definitions which is, dominant eigenvalue and stochastic matrix and give you a theorem right? So, the largest dominant or the dominant eigenvalue of a stochastic matrix is equal to 1 ok. So, to prove this, what do I have to prove? So, I need to prove two things one that 1 is an eigenvalue of this matrix of any stochastic matrix and second all the other eigenvalues are less than 1. So, that is exactly what this proof does here you can take a look at it and just to give you a heads up. So, last year I use to do this that please see the proof go back and look at the proof people never look at the proofs.

So, I used to ask them in the quiz where I should be sure that people not going to answer right? So, please when I say go back and look at the proof do that ok. So, and lastly if A is an n cross n square matrix and you have this series A v 0 A square v 0 up to an vn, then this series will converge to the dominant eigenvector of A. What does a statement mean? Let us not get into the proof right what does it actually mean ok. So, let us start with very basic stuff right what is the series actually? What is each element in this series it is a vector, it is a vector everyone gets that every element in the series is a vector?

Now what do I mean that a series of vectors converges to the dominant Eigen vector, what is convergence mean? If I keep finding the next element next element next element of this series and I keep doing this as long as I can. I will reach a value n right where n is the nth element in the series which will just be a multiple of the dominant Eigen vector is that clear? You not seem to be clear everyone gets that.

So, what do you mean by if you take a series of numbers and if I say that the series converges to 0, what does that mean? If you keep finding the next element in the series, you will hit a point n where you find the nth element of the series and it will be 0 (Refer Time: 13:20) that ok. So, we will just I will leave it at that for now. Now so stochastic matrix dominant eigenvalues the connection between 2 and the convergence theorem for a series of vectors which is A v 0 A square v 0 and so on.

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Now, let ed be the dominant Eigen vector of M where M is a dash matrix in our case it is a stochastic matrix. So, what with the corresponding dominant eigenvalue be.

Student: 1.

1 ok so given the previous definitions and theorems, what can you say about the sequence? It converges to a dash of ed.

Student: (Refer Time: 13:59).

A multiple of ed right? So, there exists an n such that the a length nth element of the series which is given by this is going to be equal to some multiple of the dominant eigenvector no, no; k is some multiple no this is not related to eigenvalues yet just wait for the next statement, then you will see the difference that this is not the do eigenvalue yet.

Now, my question is what happens from here onwards, what would be the next element in the series. How many of you say some k dash into ed? What is the other pause I do not have the other option what is the other option.

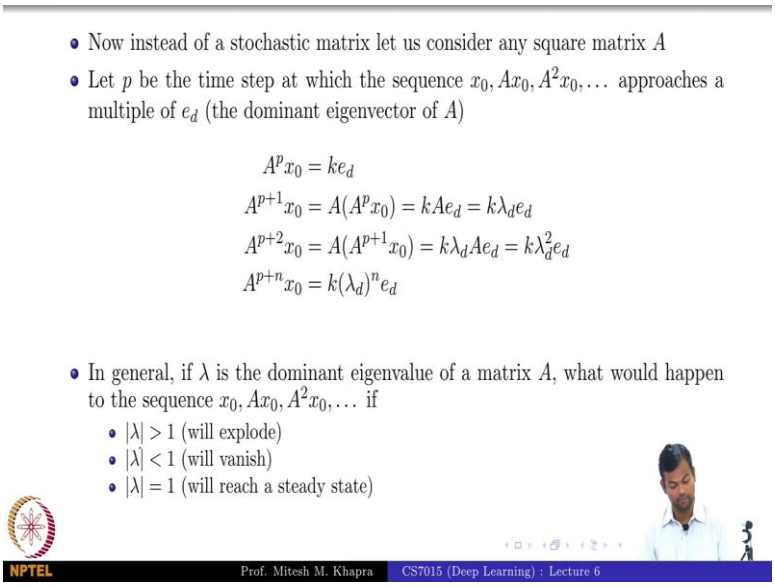
Student: k into ed.

k into ed how many of you say k into ed? A large number of ok so, you see that now just notice the eigenvalue will come up right? So, at step n plus 1 you would have M into vn

which is M into k into ed and this quantity is actually 1. So, the theorem says it will converge to some multiple of k and now if it is a stochastic matrix, what will happen after that time step? It will just remain the same vector.

So, what would happen to the number of customers in the two restaurants it will remain the same right you get that ok fine. Now, this was all for, what kind of matrices? Stochastic matrices square stochastic matrices

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But we generally care about any square matrix. In fact, we should care about any matrix not discriminate, but any square matrix will do for now. So, for a square matrix let p be the time step at which this series approaches a multiple of the dominant eigenvector.

The theorem was for any square matrix, remember it was not for stochastic square matrices. We just use this value that for a stochastic square matrix the dominant eigenvalue is 1, which it need which leads to that neat result that the num then the number of customers just becomes constant. But for any square matrix, I could write it as this that there exist some step p at which the element of the p'th element of the series would just be a multiple of the dominant eigenvector.

Now, what would happen at step p plus 1? Is this fine? what about step p plus 2, and in general at p plus k or p plus n everyone gets this. So now, can you tell me what does this

knowing this dominant Eigen value tell us about this series, when will it stabilize actually?

Student: (Refer Time: 16:25).

When lambda is equal to 1 that is the case we already saw if the dominant Eigen value is greater than 1, what would happen?

Student: (Refer Time: 16:33).

Series will explode the series will explode and if it is less than one what would happen the series will vanish ok. So, this is an important result that we will use when we are discussing exploding and vanishing gradients.

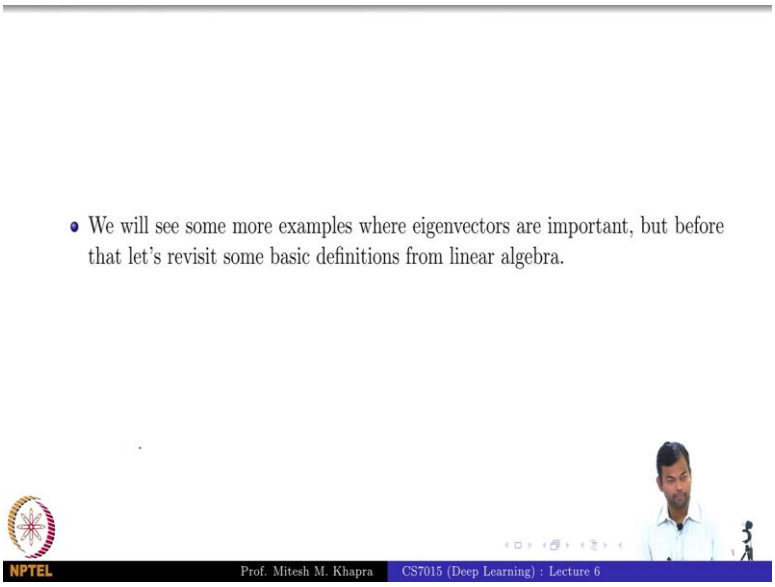
So, we will see that in the case of something one as a recurrent neural networks, you end up with something of this sort and then I will make some comments on that right? So, that is why we will be using this will come probably 6, 7 or maybe more lectures down the line ok, but we will be using it at this point. So, the main result from here is that if the dominant eigenvalue, this should be lambda d is greater than 1. Then it will explode less than 1 it will vanish and equal to 1 it will stabilize. So, that is one result one important property of eigenvalues and eigenvectors that well be needing at a later point in the course.

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Module – 6.2
Lecture - 06
Linear Algebra - Basic Definitions

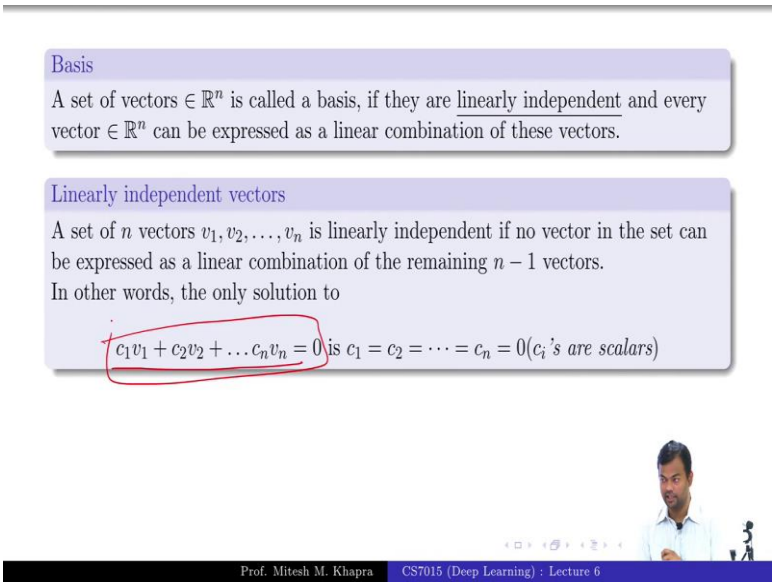
Now, from here on we will go on to something even more basic. We will start defining some basic definitions from linear algebra and these are again important for something that I need in the next lecture. So, let us start with this.

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I mean, in the process we all just see, why the eigenvectors are important for us in this course?

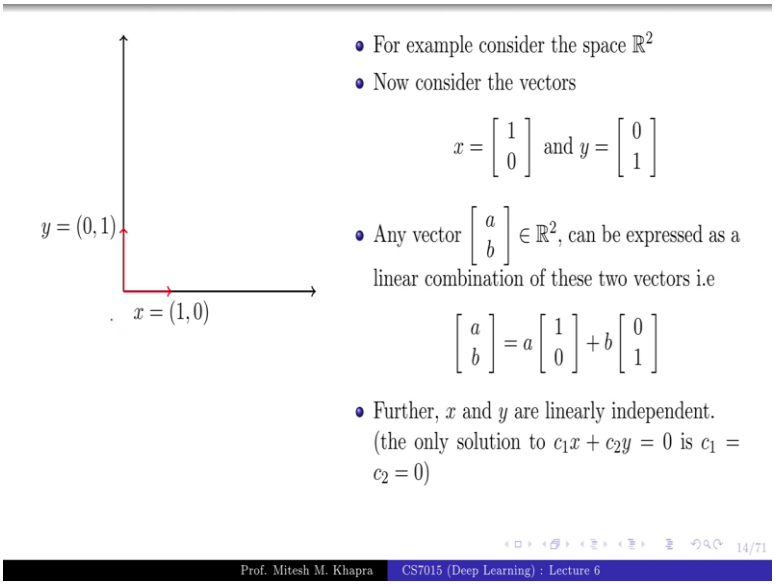
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So, how many of you know what a basis is. So, a set of vectors belonging to R n is called a basis. If they are linearly independent right, and every vector in R n can be expressed as a linear combination of these vectors. So, a set of n vectors v1 to vn is linearly independent. If no vector in the set can be expressed as a linear combination of the remaining n minus 1 vector. So, a more weird we are stating it that. So, that everyone get confuse is that if you take this linear combination.

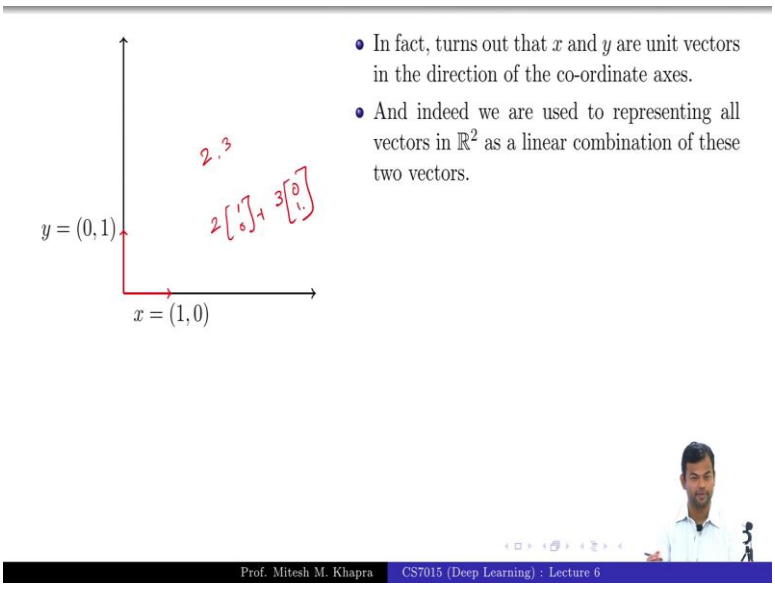
The only solution to this is all the ci's is equal to 0 and that make perfect sense right, that is that same as that linear combination, linear independence and all that thus, is make sense to everyone ok. So, what does linear independence mean that any vector from this set cannot be expressed as the linear combination of the other set; other vectors in the set and a more formal way of saying that is this. Everyone gets this what is linear independence.

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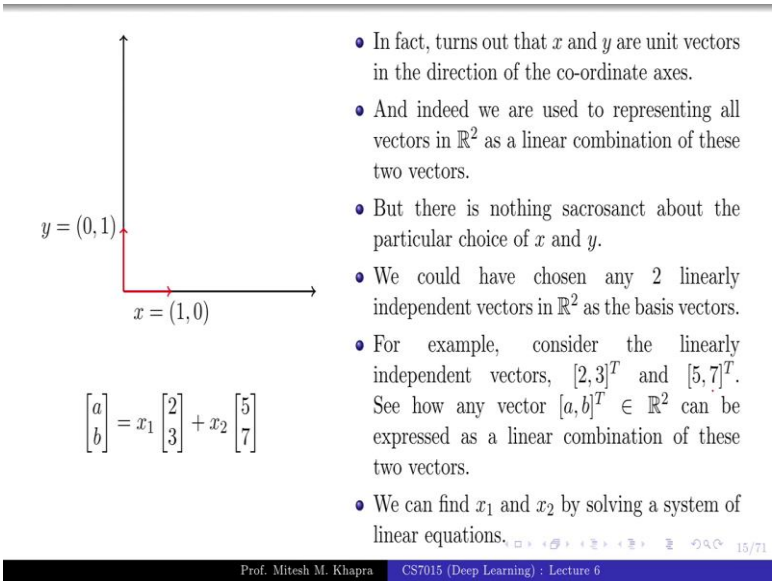
Now, let us consider some very stupid examples, again the space R 2 and we consider these 2 vectors 1 0 and 0 1. Are they linearly independent? Yes, ok they cannot be expressed as a multiple of each other right? Now, any vector ab belonging to R square can be expressed as a linear combination of these 2 vectors ok. And x and y are linearly independent the only solution is c 1 x plus oh sorry, c 1 x plus c 2 y is c 1 and y equal to 0; What about if I move to R 3 1 0 0 0 1 0 and 0 0 1. So, x y and z axis right are the unit vector.

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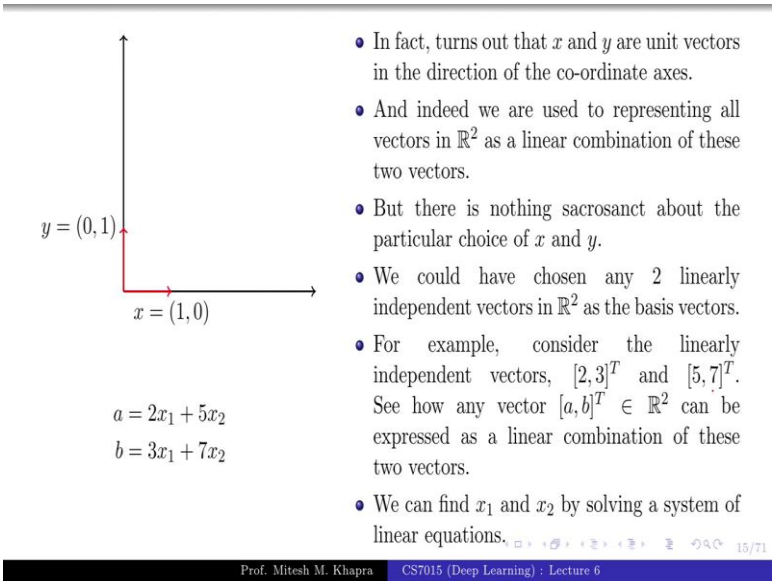
So, in that x and y turns to be unit vectors in the direction of the coordinate axis. And we are used to representing every point in R 2 as a linear combination of these 2 vector is that exactly what I what we do. So, when we say that I have a point 2 comma 3, I am actually telling you that the point is 2 1 0 plus 3 0 1 right? I am expressing at are the linear combination of the coordinate axis.

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But now this nothing sacrosanct about x and y right; I could have chosen just about any other axis. So, in particular we could have chosen this as our basis are these two vectors linearly independent can any vector and R 2 be expressed as a linear combination of these two vectors sure. So, I give you a vector a b, how do you going to express it as a linear combination of these two vectors? So, you will do it this way right? How will you find that values are the x 1 and x 2; so, other linear system of linear equations right?

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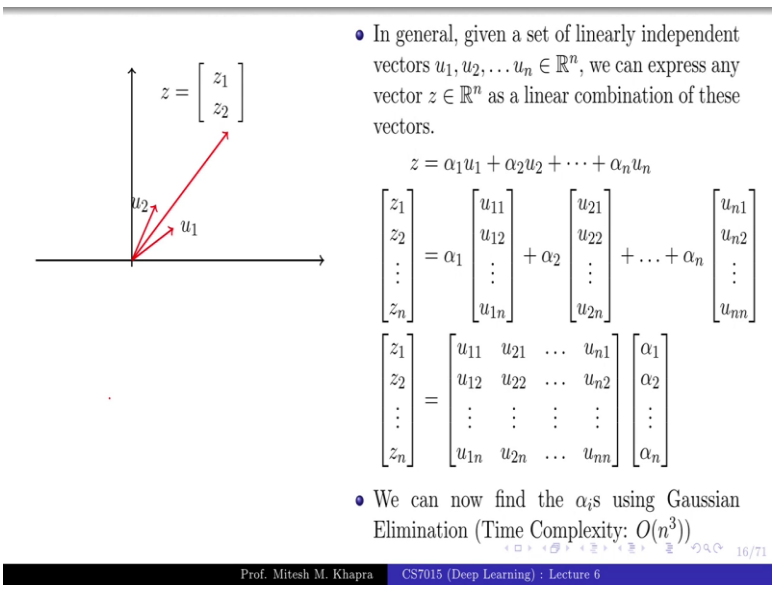


So, this is what you will do. I know all are good in doing this and what do we actually do when we do this? What is the algorithm that we use? How do we solve this? What is the algorithm that you use solving this?

Student: Gaussian elimination

Gaussian elimination right, in 2 variables of course we do not call it an algorithm. That is what we did in 8 standard or something, but when we come to engineering we call it Gaussian elimination right? So, the same algorithm.

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So, in general given a set of linearly independent vectors, we can express any vector that belonging to Rn as a linear combination of these vectors right? I can say z is equal to alpha 1, u 1 plus alpha 2 u 2 and so on given alpha 1 to alpha n are linearly independent ok. So; that means, any vector in Rn can be expressed using these vectors which form the basis of Rn does that make sense. That is why call the basis vector because, anything else these are the fundamental vectors using these anything else can be expressed in that space it is that clear.

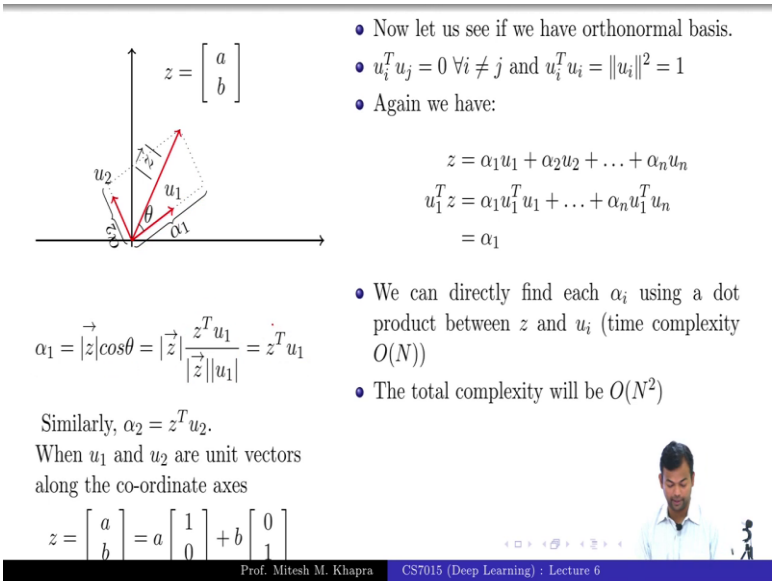
So, this is how it will be. How do I write this in matrix notation A? There are lot of these and these thing I do not really understand what you mean by that, yeah good. So, this is what you mean. So, that we writing same in matrix notation and now this is again a dash a system of linear equation there was a lot of space to fill and one dash good. So, system of linear equation and again you can solve them using.

Student: Gaussian elimination

Gaussian elimination, what is the complexity of Gaussian elimination. Let us see options right n n square n cube [FL], n cube right the Gaussian elimination the complexity is O N cube right? And I am not doing all this just to the sake of time pass I have a point of make which I will make on the next slide right? So now, this was for any basis; that means, if you have any n linear independent vectors.

Now, I will consider a special basis where instead of n linearly independent vectors. In addition, these vectors are also orthogonal ok. Orthogonal vectors are linearly independent ok. So, a set of orthogonal vectors are linearly independent, but the converse is not all this right.

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So now, let us see what if we have an orthonormal basis; that means, a basis consisting of orthonormal vectors. So, orthonormal is combination of 2 words ortho means, the 2 vectors are orthogonal and normal means all the vectors are unit vectors; that means, I am normalized them by their magnitude.

So, what is the condition that holds ui transpose uj is equal to 0? If I is not equal to j and ui transpose ui is equal to 1. Ok now, what happens in this special case, so we have this again we can express any vectors z as a linear combination of these. Now, let me try to do this I am just pre multiplying this equation by u 1 transpose. What happens on the right hand side everything disappears all of the this terms will disappear because, they are of the form ui transpose uj where i's not equal to j and the first term is.

Student: One.

One, so what remains alpha 1. So, you can directly find alpha 1 using a dot product of 2 vectors. What is the complexity of this operation n th is just n products ok? Now how many such alphas do we need to find.

Student: (Refer Time: 6:09).

n of those. So, what is the complexity n square. So, that is now you see why an orthonormal basis is a very convenient basis, you can get all these coefficients just by doing a dot product between 2 vectors and later on I will show you that you might not

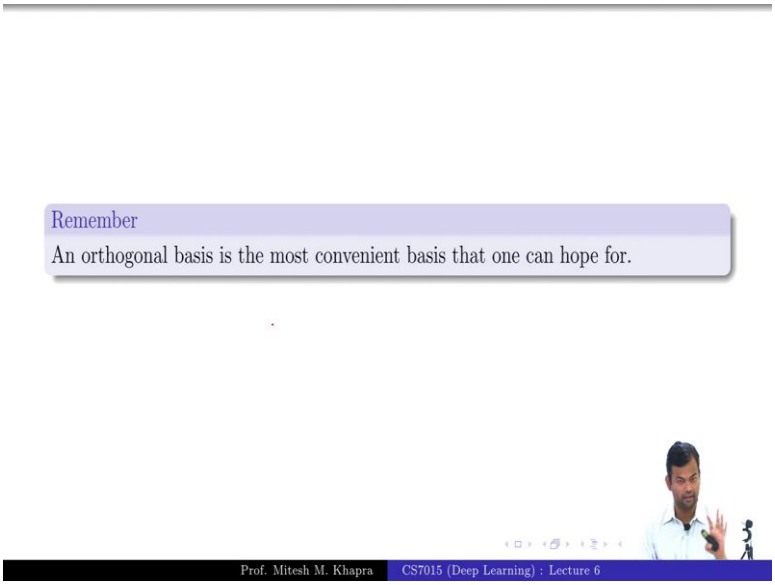
need all of these. You might just need some subset k of these right so; that means, you just do k of these dot products and get these values. So, do you now understand the meaning of what is why, why do I say it is orthonormal basis is the most convenient basis that you can hope for right.

So, the another way of looking it right at it is again just to make few more comfortable with vectors and projections and so on right? So, this was your original point z 1, z which is a comma b right? And how do you actually draw that vector that, this is a and this is b ok. So, how do you find the coordinates actually you projects on to your basis vectors which were these x and y vectors. That is how you found the components along those the coefficient along those.

Now, instead of this x and y, I have any other set of vectors which is u 1 and u 2 and I will do the same thing I will project this on to u1 ok. I will project this on to u 2 and that projection will give me alpha 1 and alpha 2 right? So now, what is alpha 1 and that sense this is z, this is alpha 1 and this is theta right? So, alpha 1 is equal to z into cos theta ok. And what this cos theta? So, again you arrive at the same thing fine.

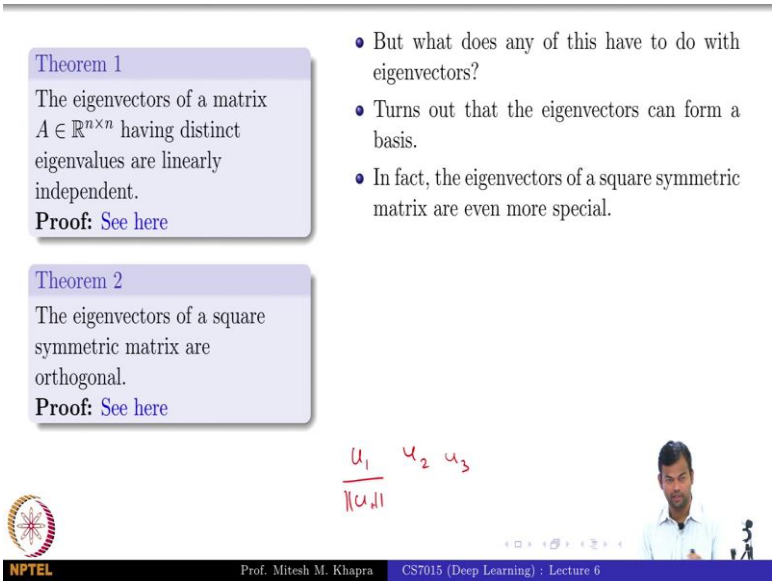
So, essentially taking a projection of a vector on to your basis is this fine to everyone. There is just to difference arriving at the same formula. That alpha is are given by a dot product between the basis vector and your original vector.

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So, an orthogonal basis is the more convenient basis that you can hope for. That is the point, which I wanted to have you are convinced about that.

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Now, but what does any of this have to do with eigenvectors. I started off with eigenvectors, I proved one property there and then I came to this linear algebra basic definitions and what a basis is set of linear independent vectors. And I eventually showed you that, an orthonormal basis is the most convenient basis that you can hope. So, what does any of this have to do with eigenvector.

Student: (Refer Time: 8:50).

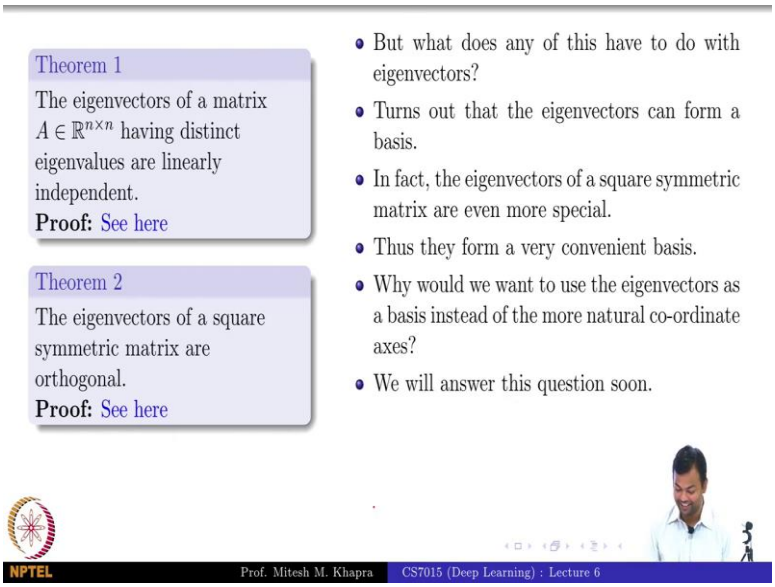
Always for us, square symmetric matrix right, why do you care about square symmetric matrix? Not sure yet. So, we get to that. So first of all, it is turns out the eigenvectors can form a basis and this is for any matrix. So, the eigenvectors of a matrix having distinct eigenvalues are linearly independent.

So, does every matrix, if I have an n cross n matrix will it have n eigenvectors? No, it can have less than or equal to eigenvector depending on the (Refer Time: 9:15). So, what is this saying is that if these eigenvectors are having distinct eigenvalues ok. Then these eigenvectors would be linearly independent, fine ok and turns out that for a square symmetric matrix that is the even more special, the eigenvector of a square symmetric matrix are.

Student: Orthogonal.

Orthogonal right and we already know that orthogonal is good right? So, remember, when we have orthogonal we do not really care about orthonormal. Because, that is it is a simple operation, if you have a set of vectors u 1 u 2 u 3 which are orthogonal, you can just divide them by the magnitudes and just get a set of orthonormal vectors right? So, orthogonal and orthonormal, I will use it interchangeably ok. And, whatever I done thus they form a very convenient basis. So, the eigenvectors of a square symmetric matrix form a very convenient basis.

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So, that is how I connect the parts which was about the eigenvectors to the second part, which was about basis. And why would we want to do this and we already we had a coordinate axis that is the very good basis 1 0 0 0 0 1 0 0 1 and n dimension similarly.

So, why should I want to use the different basis? I have said that eigenvectors is a very convenient basis, but why do I care about it. I already have a very, very convenient basis which is just these 1 or 2 vectors are along these directions right? So, why do I care about a different basis? I understand that I that is there somewhere, but something more than that that is one advantage which I will talk about. What else more interesting? Ok, in what sense I love the power which comes with my job right? That you give a right answer and still I can embarrass you know. So, that is correct actually.

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

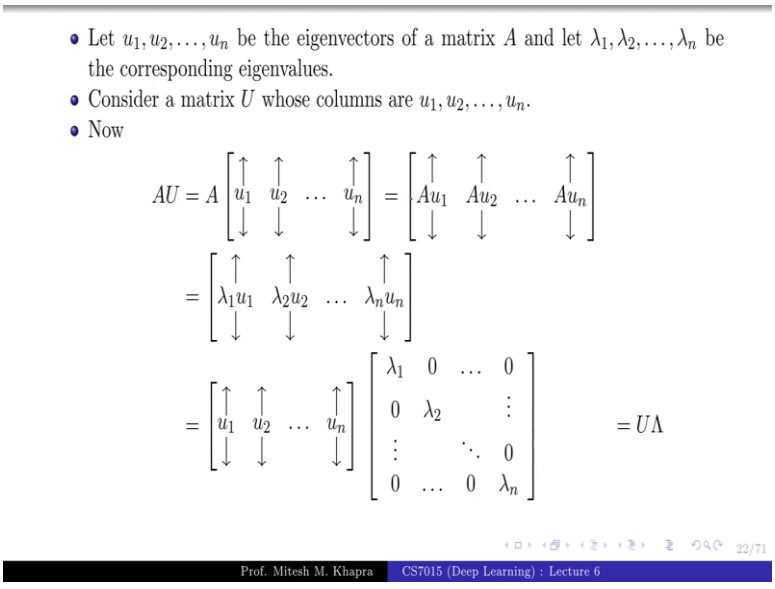
Module – 6.3
Eigenvalue Decomposition

Lecture – 06

In this module we will study Eigenvalue decomposition. So, the answer to that was actually which I was hoping all of you will give because all of you have done two prerequisite, which is linear algebra and machine learning, both of them teach you principal component analysis. So, I was hoping that you will give that answer.

Now, can you give that answer he already of course, gave that answer, is that make sense? So, we relate it to that so, but before going to principle component analysis, we look at Eigen value decomposition.

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This is very straightforward. So, let u1 to un be the Eigenvectors of a matrix A and let lambda 1 to lambda n be the corresponding Eigenvalues.

Now, I am going to construct a matrix U, such that the columns of U are these vectors u1 to un, is that fine, what u looks like. And now I am going to do this product I am taking a the product of the matrix A with the product of with the matrix U, where U is this right.

It is the all the Eigen vectors tagged one after the other is this fine. The next step I am just pushing the matrix inside. If you know the 4 different ways of multiplying a matrix you will know that this is correct. Or else for now just thing that you can just push the matrix inside.

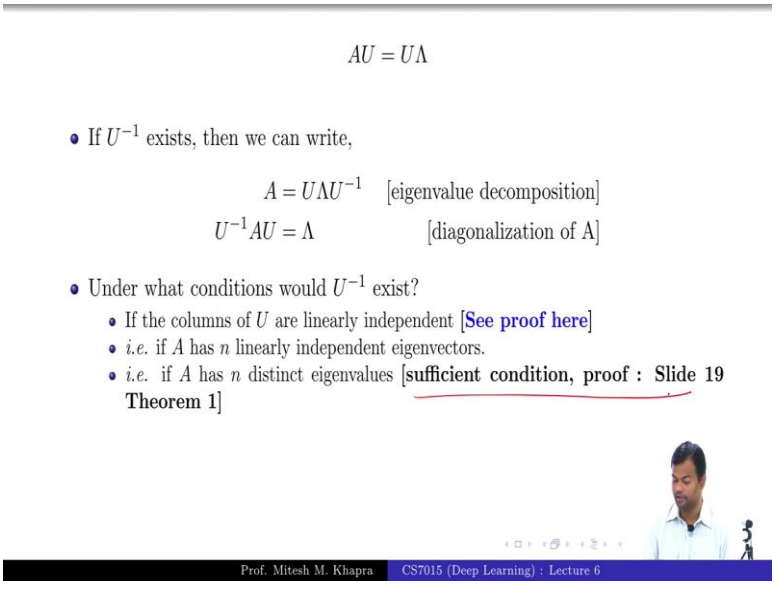
Now, what is this I can replace them by the lambda 1 u 1 lambda 2, because a u 1 is equal to lambda 1 u 1 by definition ok. Now can you write this again as a product of 2 matrices, one is of course, the matrix U and the other is?

Student: Diagonal.

Diagonal. So, the diagonal matrix will come first or the matrix U will come first? How many if you say U will come first? How many if you say the diagonal matrix will come first? The sum is never one ok.

So, it is going to be like this ok. And you can write this as U lambda. So, U is again the vector the matrix containing the Eigenvectors of A and lambda is a diagonal matrix where every diagonally element is a corresponding Eigen value.

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Now, this is what we have so far A into U is equal to U into lambda. Now suppose U inverse exists, I will assume that U inverse exists and later on I will tell you under what conditions it exists, then I could write it as this any of these 2 forms in one case. I am post multiplying by U inverse in the other case I am pre multiplying with U inverse ok.

So, this is known as the Eigenvalue decomposition of a matrix. And the other way of writing it is known as diagonalization of the matrix right. You take a matrix apply some operations to it so that, the result is a diagonal matrix is this clear to all of you is very straight forward ok. And again Eigen vectors play an important role in this. Now the important question is under what conditions would U inverse exist. U inverse would exist if the columns of the matrix U are.

Student: Linearly independent.

Linearly independent, ok. Do we know the columns of the matrix are linearly independent?

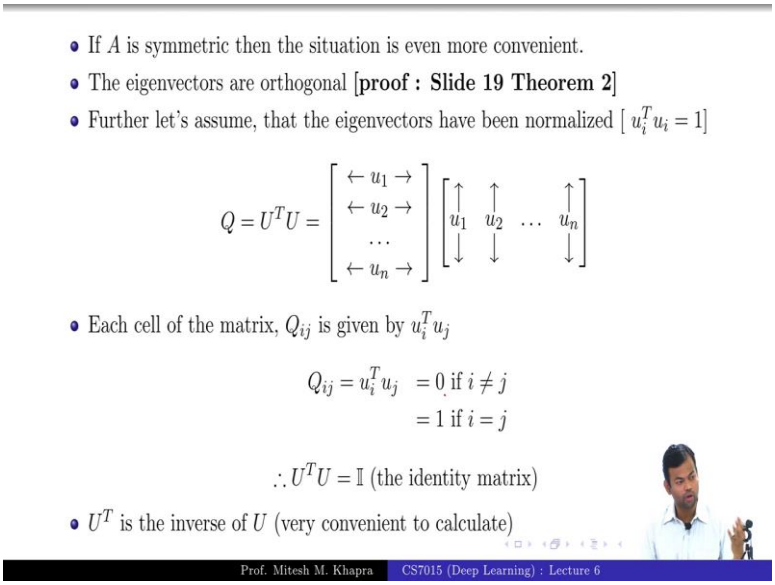
Student: Yes.

Yes, because it is a.

Student: (Refer Time: 03:10).

Set of Eigenvectors and we already saw the proof that the Eigen vectors are linearly independent ok. This just follows whatever I say ok. Now do we need proof for this I slide 19 we did this, I did not realize it fine.

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Now if A is symmetric the situation is always more convenient, why is it?

Student: (Refer Time: 03:40).

What would U be?

Student: Orthogonal matrix.

What is an orthogonal matrix actually?

Student: (Refer Time: 03:47).

So the Eigenvectors are orthogonal. So, we have this situation right. Suppose I want to do U transpose U ok. This is how that operation would look like ok. Now what is the ij'th entry of the resultant matrix ?

Student: Dot product.

It is the dot product between the.

Student: ui and uj.

ui and uj. Everyone gets this right, the ijth entry of this product is going to be the dot product between ui and uj. This dot product would be dash if i is not equal to 0 or j.

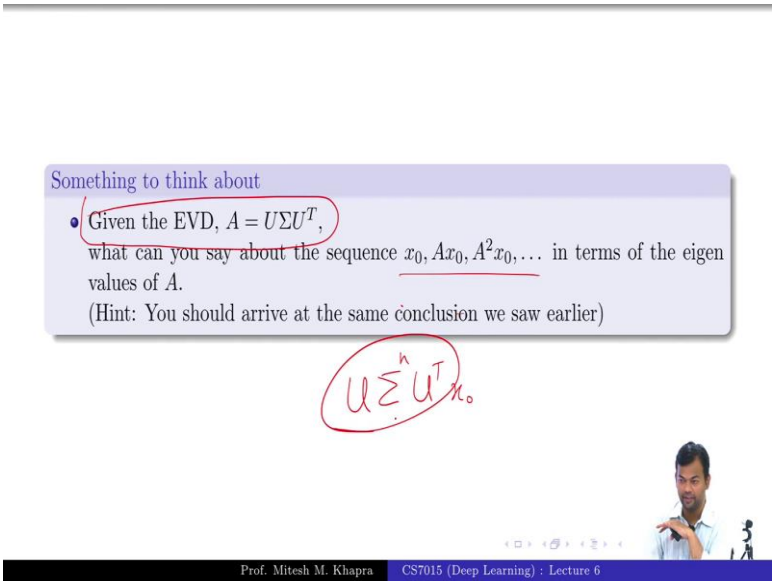
Student: j.

J and there is no point in this. So, each cell of the matrix Q ij is given by the dot product and it is going to be 0 if i not equal to j, and it is going to be 1 if i is equal to j ok. So, U transpose U is equal to the identity matrix; that means, U transpose is the dash of U.

Student: (Refer Time: 04:45).

Transpose of U and of course, inverse also ok. So, U transpose is the inverse of U. And it is very convenient to calculate what is the complexity of inverse? So now, you appreciate that that is a that has high complexity and in this case if the vector if the matrix is orthogonal; that means, it is a collection of orthogonal vectors and the inverse just comes for free right.

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So now given this situation, and do not read the hint as if this is going to help, but yeah, what can you now say about the sequence? The same sequence that you saw earlier. So, I have given you that the EVD of a is equal to U sigma U transpose, where u is the collection of the Eigenvectors and sigma is the Eigen values the diagonal matrix containing the Eigenvalues.

Now, what given this and ignoring the knowledge of the first section of this lecture can you tell me something about this series? What would be the nth element of the series?

Student: U sigma power n.

U sigma.

Student: Power n.

Power n.

Student: U transpose.

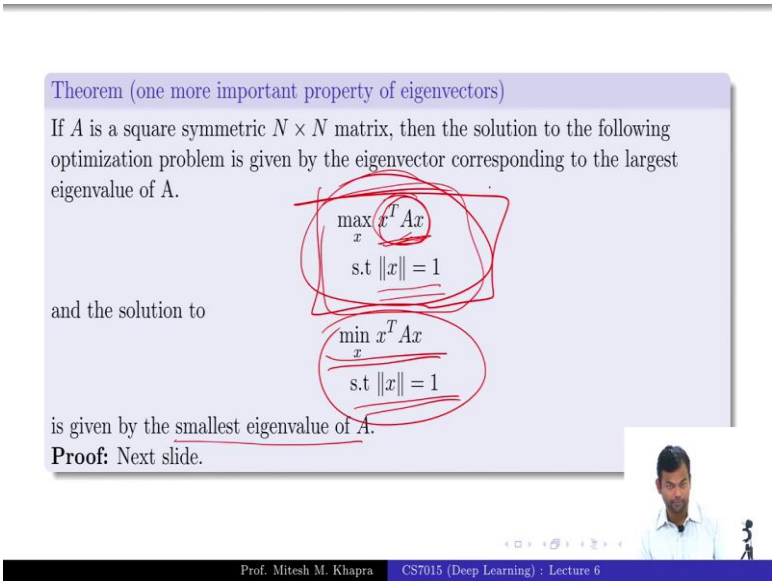
U transpose and you arrive at the same conclusion right? Where I was talking about this operation right. So, if we can say something about this matrix then we can say something about this series what can you say about this matrix, if the largest Eigenvalue is greater than 1 as you keep raising it is power that value is going to explode. And hence, the

entire product is going to explode less than 1. That product is going to vanish and everything else would be less than that right remember is the dominant Eigen value.

So, everything would be less than that. So, that product will vanish ok. So, the same conclusions you can arrive at right. So, that is why I want to do these sections again. So, you would have done these in linear algebra, but you would have not arrived at these conclusions from a very different interpretation, but I want to focus on the interpretations that I care about. I do not, how many of you have seen this series in the course on linear algebra? You have ok, but I do not see why anyone else would teach this is not required is only required for some things that I want to do in the course right, that is why; I wanted to do this section.

So, everyone is comfortable with Eigenvalue decomposition it is a very simple stuff right I mean there is no proof or anything involved there we just use some properties of Eigenvectors and Eigenvalues and do it.

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Now, there is one more important property of Eigenvectors, which well use today. So, let us see what this means right. You have a matrix A which is an n cross n matrix ok. And your import interested in computing this value, x transpose A x where x belongs to Rn x belongs to Rn.

So, what am I trying to do here of all these vectors possible in Rn, I want that vector which maximizes this quantity. What is this quantity scalar, vector, matrix, tensor?

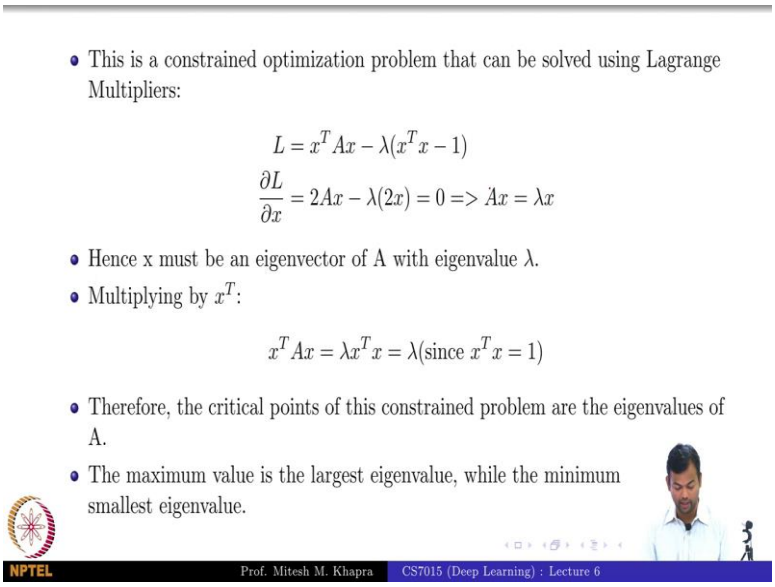
Student: Scalar.

Scalar ok, such that x is equal to1. This is the problem that I have been given to solve why it is not clear as of now, but suppose this is a problem I am trying to solve, or the inverse of this which is minimize the same thing, of all the vectors in Rn find the vector which minimizes this quantity, subject to these constraints. Then the solution for this is given by the smallest or largest the solution is the smallest Eigen value of A.

And x is the Eigenvector corresponding to that. So, if you are trying to minimize and the solution is a smallest Eigenvalue, we need to clarify that if you are trying to maximize and the solution is the largest Eigenvalue is that clear and the value of x would be the corresponding Eigen value. So, largest Eigen vector is the same as something that we have defined today dominant Eigen vector right?

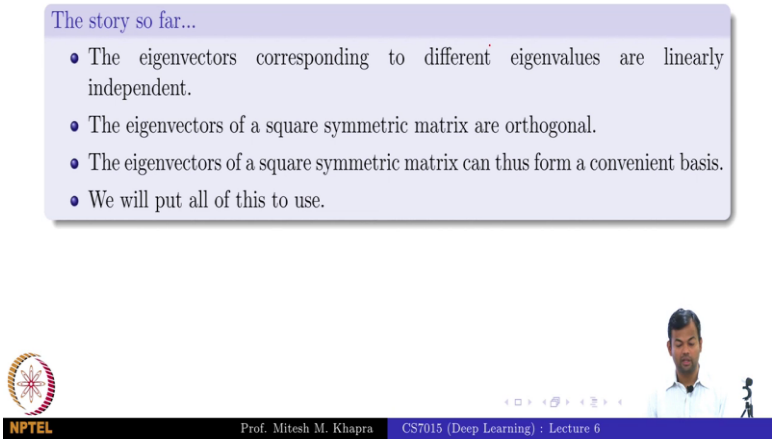
So, let me just repeat. So, that there is no confusion. Let us focus on this problem. The solution to this problem that is the x which will give me the maximum which will maximize this is the dominant Eigen vector of the matrix A right, is that clear? Fine ok. And if you want to minimize it is going to be the smallest Eigen vector; that means, the inverse of the dominant.

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So, there is a proof for that I will not go over the proof you can take a look at it at your own leisure.

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So, what has been the story so far. The story has been that the Eigenvectors corresponding to different Eigen values are linearly independent.

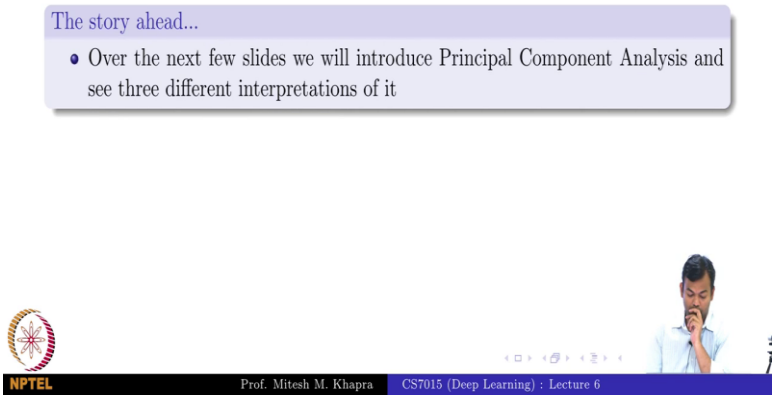
If you are dealing with the square symmetric matrix, which is something that we will deal with soon. Then things are even more convenient because the Eigen vectors are actually orthogonal ok. And they form a very convenient basis, and now we are going to put this to use when we talk about principal component.

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Module - 6.4
Lecture – 06
Principal Component Analysis and it is Interpretations

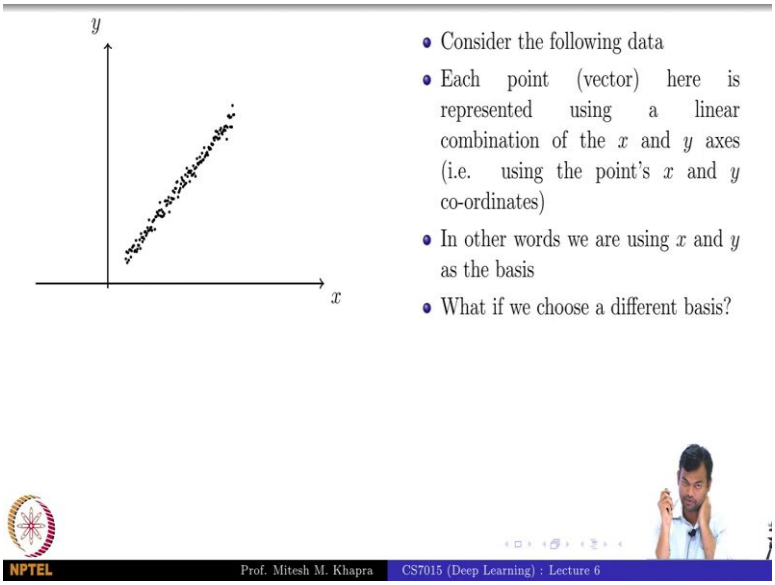
So, in this module we will talk about Principle Component Analysis and it is different Interpretations. In this model we will look at one interpretation and then in the rest of the module some other interpretations.

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So, the story I add is going to be this we will talk about PCA and it is Interpretations, ok.

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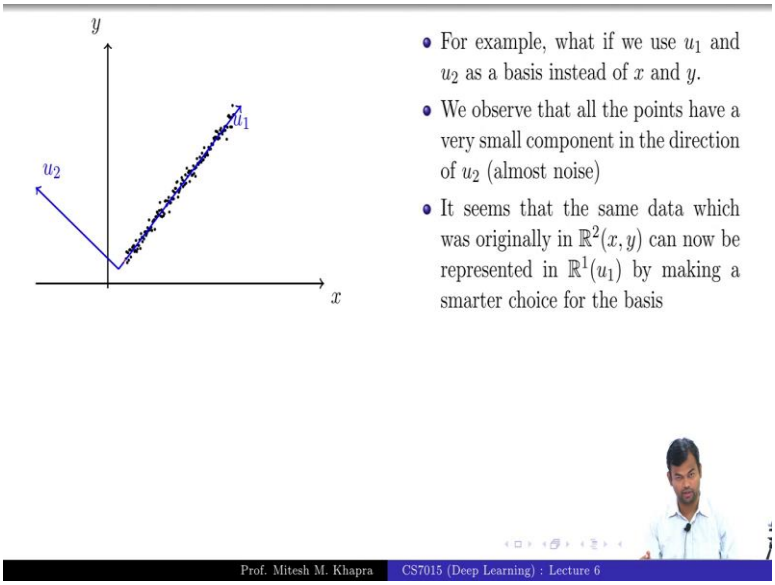
So now let us try to motivate PCA first consider the following data ok. In what dimension is this data?

Student: 2 dimension.

2 dimensions it is R 2 ok. And each point here is represented as it is x coordinate and using it is x coordinate and it is y coordinate ok. Now it means that were using x and y as the basis right? That is clear that is the standard way that you would do any data point you will just represent using that basis.

Now, what if we choose a different basis? Let me give you 1 basis and then let me ask you some questions on this.

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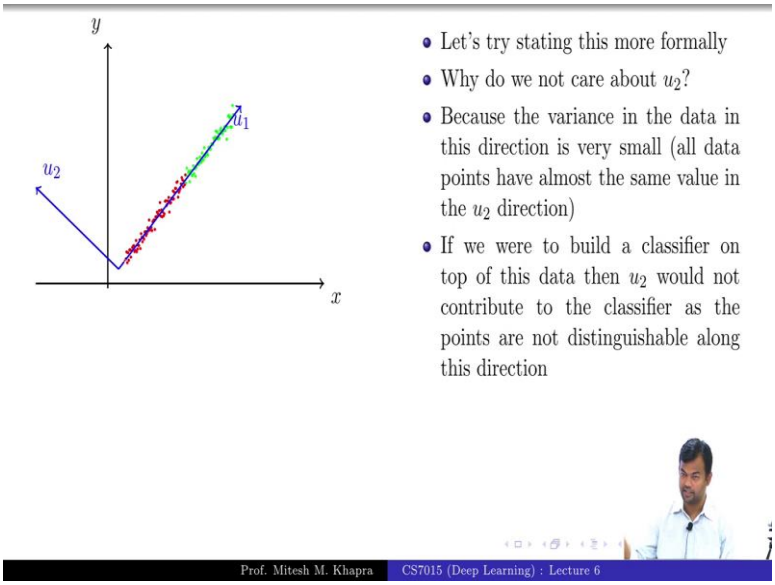


Suppose we chose this basis. So, in the previous modules we made a case for the x and y coordinate axis, there is nothing sacrosanct about it you could use any basis; The only condition on the basis that the vector should be linearly independent and in fact, if they are orthogonal it is even better right?

So now I have given you a different basis now what do you make any observation here? So, they have all the points here have a very small component along the u 2 axis right? So now, this so far this point right, if I consider at this point then this is the component along the u 1 axis. So, that is it is u 1 coordinate as akin to the x coordinate and this is it is u 2 coordinate akin to the y coordinate, is a are the arrows clear here.

So; that means, there u 2 coordinate is very small, and it is also very small for all the data points right? So, it is almost as if there is some noise there it is all within some epsilon. Now so it seems that the data which were actually represented in R 2 can actually be represented in R 1 by getting rid of this noisy dimension right? So, if you had chosen a different basis, you realize that with just one dimension you could have captured everything that was there in the data, and the other dimension was just adding noise it was redundant there is hardly any information there.

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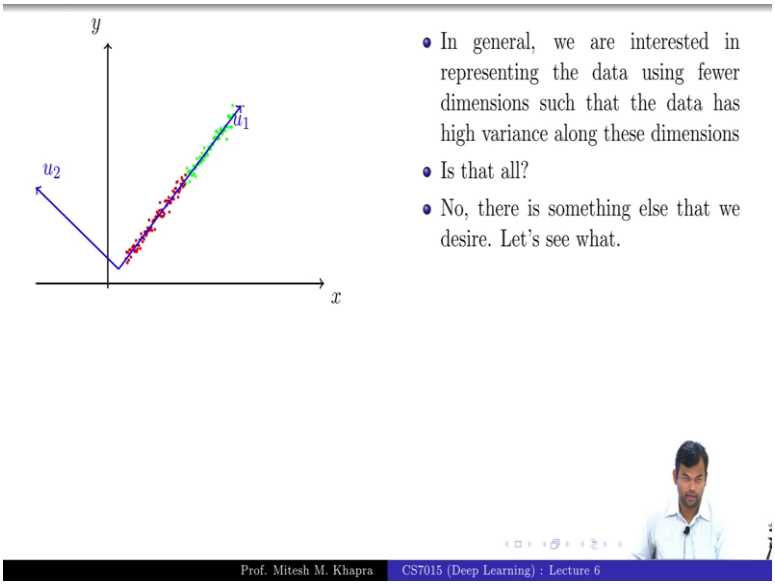
So now can you state this more formally because this is this intuition, but can you stated more formally in terms of things that you have learned and say probability for it for example, what is wrong with the direction u 2? The spread of the data points along the u 2 direction is very small, what is the spread mean the variance right? So, we do not care about u 2 because the variance in the data along this direction is very, very small ok. And in particular right if I were to build a classifier, then would u 2 have any predictive power? Because along this dimension the points are indistinguishable ok; so, think of it that you are trying to find out whether you have. So, you have say 100 candidates and you want to decide whether they would be good basketball players or not.

And quite naturally all the people that have shown up are say 6 foot 2 and 6 foot 3 inch and so on; and there in a very small height difference between them and all of them are 6.2 is the average and very close the spread is not much. So, this feature is not going to help you decide whether this person is going to be a good basketball player or not. You will have to rely on other features where the variance is more for example, how many teams has he participated in the past, how many matches as he won as a team as a member of some team and so on it.

So, those who expect some spread to be there all these 100 candidates might have different things right? But if the height is the same for all of them it is not going to be a

good predictor. And that is exactly what is happening along the u 2 direction. The points are almost indistinguishable there that is why; it does not matter.

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So, in general given any data, now this was a simple case where the data was R 2 I am talking about the general case where the data is Rn right, and you will find this situation in higher dimensions also. So, you would not want to use that entire n dimensional data where you know that there are some columns, along with the variance is very small. So, you want to represent the data with fewer dimensions, such that; the data has high variance along those dimensions.

Now, let me just clear a confusion here right? So, I am not saying that take your n dimensional data ok. Find the variance across each of these dimensions and then throw away the columns which have the lowest dimension, in this particular example if you had done this, what would happen? Could you have done that think of the original dimensions x and y. Along, these two dimensions there is enough various in the data, right? The x coordinates vary from here to here. And the y coordinates also vary from this point right up to that point right? So, there is enough spread in the x and y coordinates.

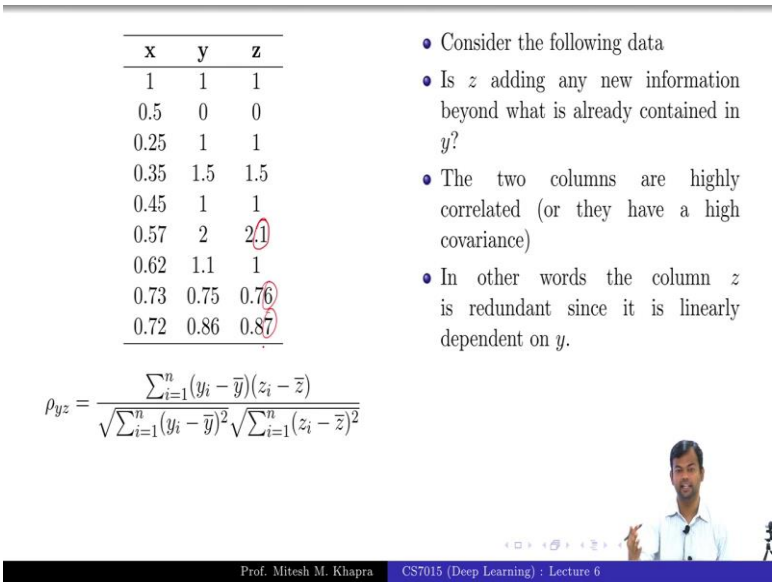
So, in your original data I am not saying that pick look at each column, and see if there is no variance along that column then throw it away that would not work because you might end up with the situation that there is enough variance across each of these

dimensions. It is just that when you look at the data from a different angle; that means, you projected onto a different basis this becomes clear right?

So, you see the difference I that is not the same these two things are different operations. So, what I am looking at is projecting the data to a different basis, that is exactly what I did with u 1 and u 2. And then some things became clear about the data. Now this projection along a different basis, I would be interested in doing that only if; I can get rid of the number of dimensions right? If now I had already had one basis where I had n dimensions. Now if the new basis is also going to be that all these new n dimensions that I have come up with are important then you are not gaining much, I do still have this high dimensional data, but you would like to project it in a way that you get rid of the lower variance dimensions.

So, you might project it onto n dimensions, but you want to rank these dimensions according to variance and then throw away some of these dimensions, is that clear; is the objective clear? Ok fine. Is that all that we care about; n dimensions' project to a new basis and throw away the key dimensions which have less variance, is that all? What else would you want? People have done the MLPR course, no I would not. So, I am not going to classification or anything I just want a better representation of the data at this point. I am not really thinking about what I want to do with the data, maybe you are talking in terms of classification. And we have already seen even if the data is not linearly separable we have solutions for dealing with right? So, that is not a critical point ok. So, there is something else that very interested in and let us look at that.

(Refer Slide Time: 06:49)



Now, consider this data I have 3 dimensional data ok. Do you find something odd about this data?

Student: (Refer Time: 06:56).

y and z are.

Student: (Refer Time: 07:00).

Are highly dash.

Student: Correlated.

Correlated right do you want these dimensions? Can you think of any practice such dimensions occurring? Height in centimeter and height in inches, someone would have just given you data right? Or if you if you take the credit card a credit card fraud detection case right? Someone would give you the salary and it would also give you the income tax now these 2 are highly correlated right?

So, then you do not really care if you have one you could probably almost with certainty predict the other right? Modulus some rules right because you get some tax exemptions and all that, but still. So, you can have this in practice, but even in our oil mining case your salinity pressure density those things could be related right? So, z is not adding any new information beyond what y is happening. So, the 2 columns are highly correlated.

So, actually yeah this is the formula for correlation, all of you know this anyone who does not know this formula good. So, not nothing is a stupid question right? So, you can always ask.

So, y hat is the mean of this column ok, sorry y bar; z bar is the mean of this column and this is how you compute correlation this is just the formula ok. So, from every entry you subtract the mean ok. So, this is known as centering the data. So, if you do this what would the mean of the new data be?

Student: 0.

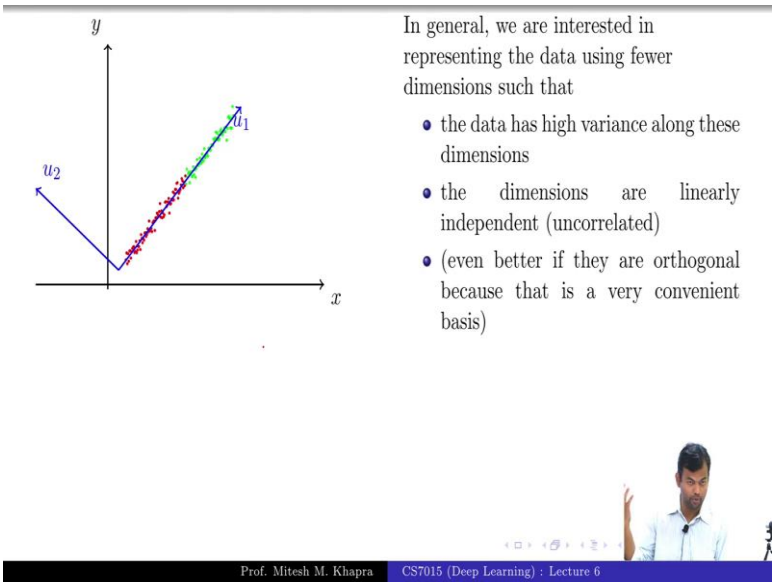
0 right? So, that is why it is called centering the data ok. So, I will have 0 mean 0 mean and you. So, what does this; what is the intuition behind this formula? Does anyone know? Can anyone tell me? So, this is a summation ok. So, this quantity is going to be high if the summation is high. It is a summation of some n terms now these terms could be positive or negative; If all the terms are positive what would we happen to the sum?

Student: (Refer Time: 08:50).

It would be high if there are some terms which are negative it would be low. Now when would all the terms be positive whenever y is above the mean, z is also above the mean right therefore, this quantity is positive this quantity is positive. Whenever both are below the mean again the product would be positive. When one is above the mean the other is below the mean, then there is something wrong happening right and in that case you will have a negative term right?.

So, for more details of course, you can refer your other textbooks and so on, but this is just the intuition an important step here is to 0 mean the data right? We are computing the subtracting the mean of the data. ok Another way of saying this is that the column z is actually linearly dependent on y ok. It is almost linearly dependent I of course, have some noise 2.1, 0.76 and so on, but it is largely linearly dependent, I can get I can write z as some c times x fine ok?

(Refer Slide Time: 09:48)



So now can you tell me the refined goals that we have? We are interesting the representing data using fewer dimensions such that; remember that when I say fewer dimensions I mean a new set of dimensions right? It is not throwing away dimensions from the current data. We are looking for a new set of dimensions. What are the conditions that we want from these new set of dimensions?

Student: (Refer Time: 10:12).

One there should be high variance along these dimensions the new dimensions, and?

Student: (Refer Time: 10:15).

The dependence are linearly independent or uncorrelated ok fine.

And even better of course, if they are orthogonal, why?

Student: (Refer Time: 10:26).

Because we are looking for a new dash.

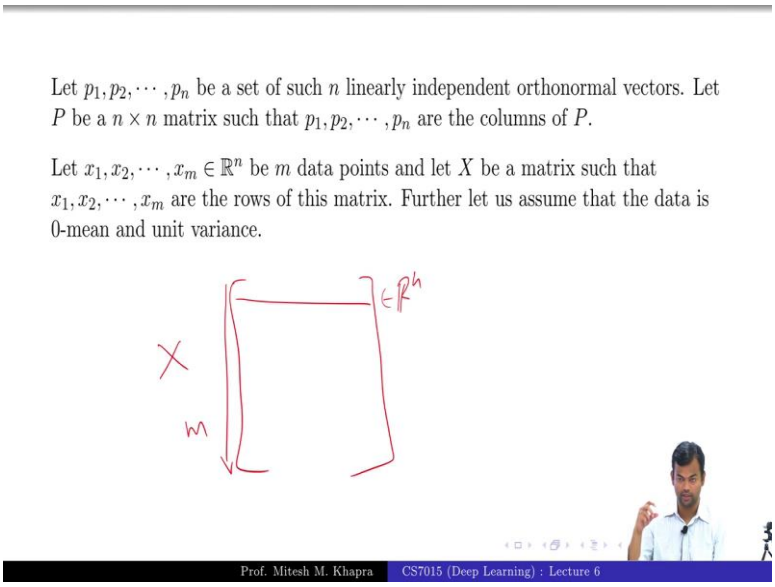
Student: Basis.

Basis and the most convenient basis is.

Student: Orthogonal basis.

Orthogonal basis, ok, fine.

(Refer Slide Time: 10:33)



So, now let us assume someone has given us this new basis ok. And let us call this p 1 p 2 pm. So, instead of this x y z and so on, someone has given us this new basis eventually we will of course, figure out how to find the basis, but let us assume that someone has given this new basis right? And they are both linearly independent and actually it is redundant actually. So, yeah this example of a redundant feature such an, orthogonal vectors is sufficient, they are linearly independent.

Let p be an n cross n matrix such that p 1 p 2 p n are the columns of p right; Same thing as we had put the Eigen vectors in a column and probably I have unknowingly given out the solution, but ok. And let x 1 to xm be the m data points given to us ok. So, we are given this data as usual we have this X matrix each one of them belongs to Rn. And we have m such data points right that is the standard thing that we are operating, and you always write this as a matrix ok and we have already done the data is 0 mean and unit variance.

Actually unit variance is not required, but the data is 0 mean fine that we will sorry I am going to deal with covariance, as a unit variance is not required. So, the data is 0 mean is what I am going to assume, but what if the data is nonzero mean I can always make it 0, mean right?

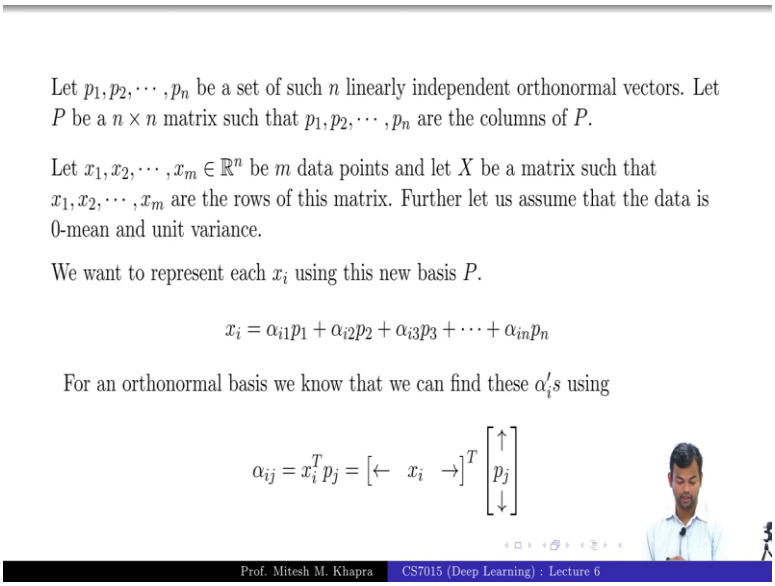
So, just to remember this is an important trick that you will always have to use whenever you are doing any large scale machine learning. So, whenever you are participating in Kaggle competitions almost the first thing that you do is standardize the data; that means, make it 0 mean and unit variance. So, why is that important?

Student: Scaling.

Right scaling issues would not be there right? So, if I have something in centimeters and some other unit in kilometers right? Now remember that always you are doing somewhere this linear operation w transpose X. You might add a non-linearity on top of that, but now if your xi dimensions some of them are in the range of 0 to 10000 some of them are in the range 0 to 10 right?

Then there is some abnormality here right, some dimensions are winnings in terms of their magnitude and some dimensions are losing out right? That is why you always make it unit variance and you also make it 0 mean you center the data ok. So, we will assume this and if we all understand if the data is already not 0 mean and unit variance we can always make it 0 mean and unit variance, just scale it and make it centered.

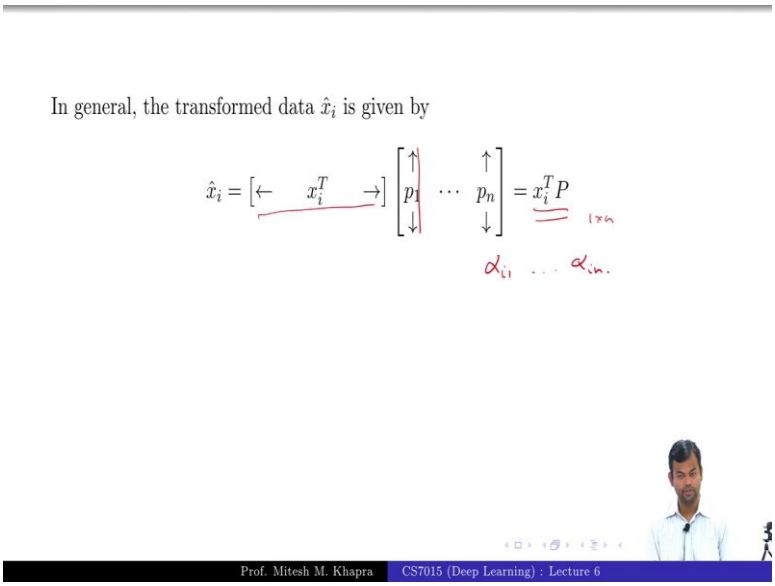
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Now, we want to represent each xi right? So, xi is one of these data points that we had; that means, one of the rows of our matrix, ok. And you want to write it as a linear combination of this new basis.

So, if you have any basis any vector you can write it as a linear combination of that basis , is it fine? So far it is ok, ok. Now for an orthogonal basis we know that we can compute these alphas just by taking a dot product of the vector with the dimension ok, and just repeating some of the things right.

(Refer Slide Time: 13:25)



So now let us see what this means; for one of the dimensions this is my data point xi which I want to transform ok, for one of the dimensions I just had to take the dot product with that dimension and this will give me how many values; One value; that means, the coordinate along p 1. I want to do it for all the n of them I can write it as this vector matrix multiplication right, what is the dimension of this?

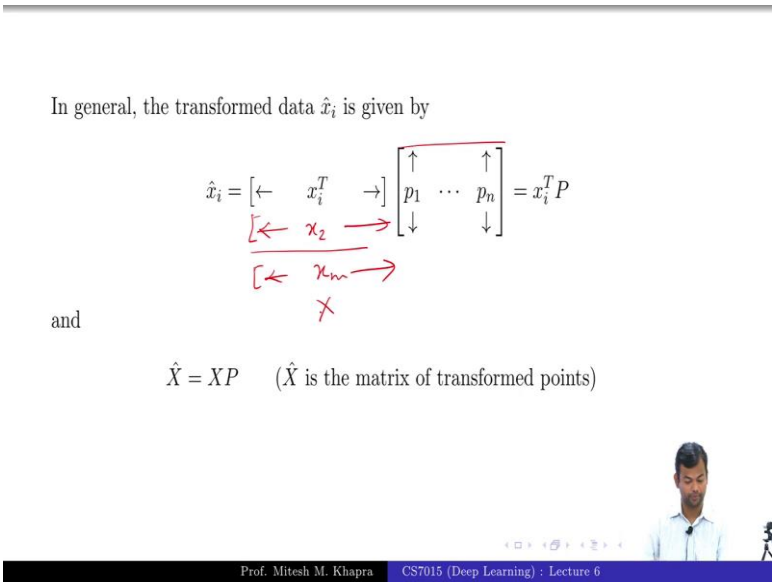
N cross 1, how many if you get that? Ok so this oh not many why?

Student: (Refer Time: 13:55)

1 cross n fine that is fine yeah how many of you get this; Ok fine yeah. So, this will give me all the n alphas is that clear for this data point.

So, it will give me alpha i 1 to alpha i is it.

(Refer Slide Time: 14:11)



Now, I want to do this for the entire data right? So, I have done it for x 1 I also want it to be done for x 2 and all the way up to x m, for each of these I would have such an operation where I have a vector multiplied by this matrix. If I just stack all these vectors I get back my matrix x. And the whole operation I can write as x into p ok is that clear to everyone ok, what is the dimension of x into p?

Student: m cross n.

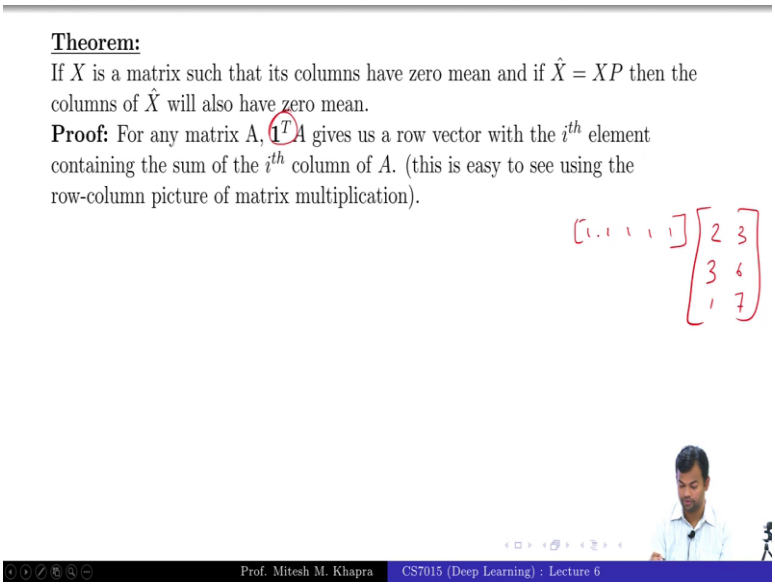
m cross

Student: n.

n right so, for all the m data points I have alpha 1to alpha n is that clear anyone who does not understand this?

So, X hat is the matrix of the transformed points is that clear? I have now the new coordinates instead of the original coordinates according to the coordinate axis. I have the new coordinates in this matrix.

(Refer Slide Time: 15:09)



Now I will just go through some very simple theorems or rather results. And I will not prove them you can prove them on your own, or other proof is there in the slides we can look at it later on right? So, if X is a matrix such that it columns have 0 mean and if X hat is equal to XP, then the columns of X hat will also have 0 mean. Is this obvious to most of you, not really is it, how many of you think it is obvious? Ok then let me just go over the proof.

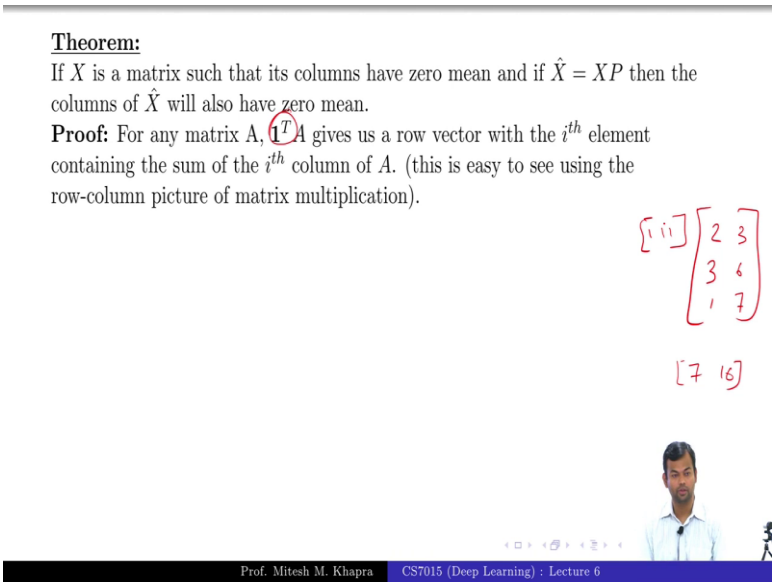
So, for any matrix A, 1 transpose A right so; that means, you have this vector this is a vector or a matrix; yeah this is a vector right? So, I have a vector of n 1. So, 1 this is nothing but a vector of n 1s. So, what is this product actually going to give me?

It will give me a vector containing n elements, what is each element?

Student: Sum of that column.

Sum of that column right, is this fine? Ok this is very obvious to see from if I have this suppose I have 2 3 1 and 3 6 7 ok, and then of course, the corresponding.

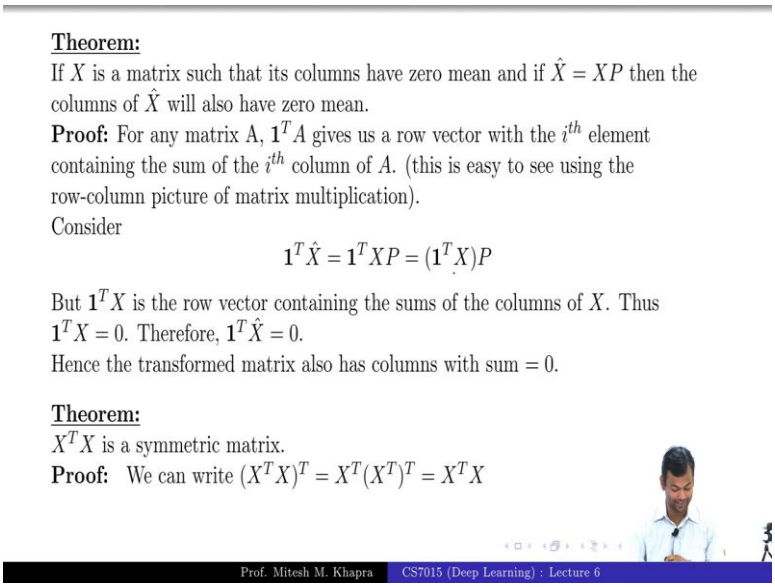
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So, if I do this multiplication I will get a 2 dimensional output which would be just 7 and 16 right? So, that is just the sum of that column.

Student: (Refer Time: 16:33)

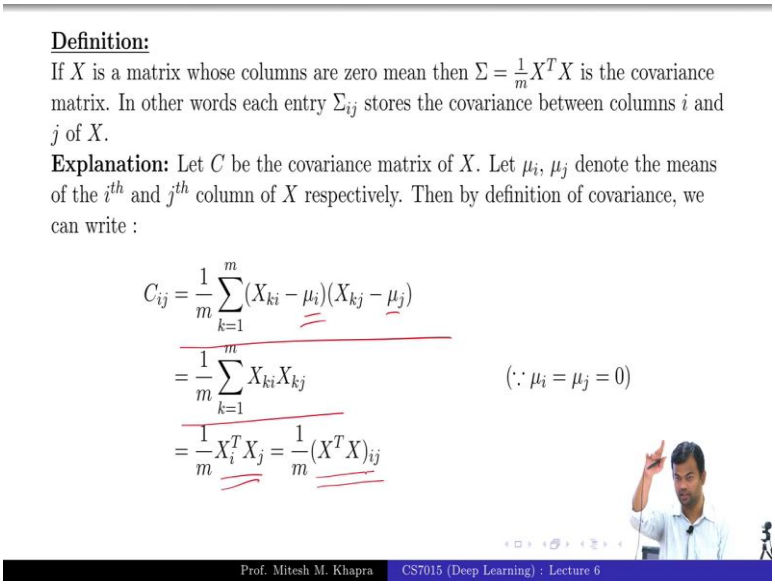
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So now, we have this X hat that is the transform matrix. Now let us see if I do this operation I X hat what happen. I can write it as this I can club it as this, what is this? It will be all 0s because the original matrix was mean 0; that means, the of the elements of all the columns each column independently was 0; that what this is going to be a zero

vector. So, zero vector multiplied by any matrix is going to be 0. Now is it obvious I hope this is obvious X transpose X is a symmetric matrix, I still have the proof for that.

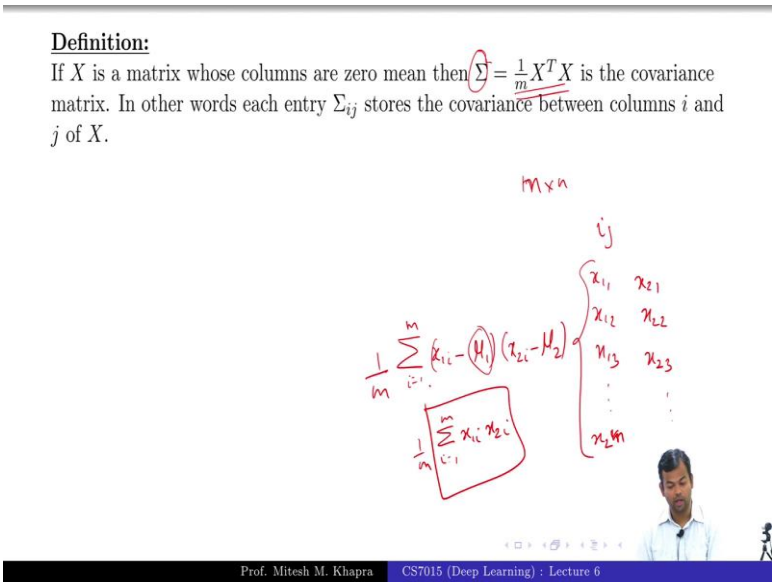
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Now, if x is a matrix whose columns are 0 mean. Then a matrix sigma which I am going to call as a covariance matrix, which is given by this is actually the covariance matrix. How many of you agree with this; how many of you have seen the covariance matrix before? So, all of you agree that this is the covariance matrix if you do not please raise your hands; If you do not you will not understand the rest of the stuff now you have to be given the right incentives.

So, let us see be the covariance matrix of X. Now what is the covariance matrix actually first of all tell me that? If I say that I have an n cross n matrix x.

(Refer Slide Time: 17:45)



Let me not make it any cross n, let me make it m cross n ok. What does the covariance matrix actually capture; what is the dimension of the covariance matrix first of all?

Student: n cross n.

n cross n ok, and what does each entry of the covariance matrix capture the covariance between the I'th column and the j'th column.

Student: (Refer Time: 18:09).

So, the entry ij of the covariance matrix captures, the covariance between the i'th column and the j'th column is that fine. Now what is the formula for covariance suppose I give you 2 columns right let us see I have give you x 1 1 x 1 2 x 1 3 and x 2 1 x 2 2 x 2 3, can you give me a formula and of course, I will go up to k or rather m.

So, what is the formula summation?

Student: (Refer Time: 18:52).

i equal to 1 to.

Student: m.

m.

Student: (Refer Time: 18:52).

Mu 1.

Student: (Refer Time: 18:53).

Mu 2 anything missing?

Student: By m.

By m anything else in the denominator? No, no is it fine ok. So, an what is mu 1? Mu 1 is just an average of this ok. So, this is the covariance formula now if the mu's are 0, then what does this boil down to?

Student: (Refer Time: 19:14).

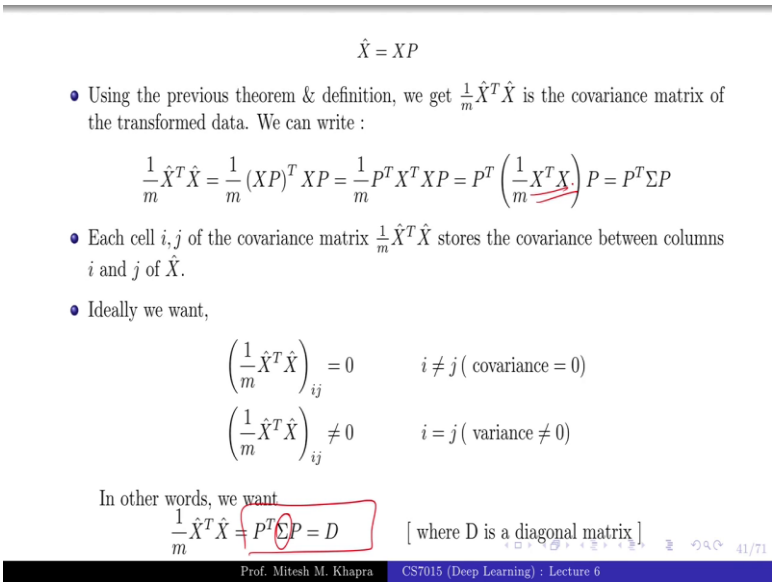
x 1 i into x 2 i , what is this quantity actually?

Student: (Refer Time: 19:22).

This is the dot product between the i'th column and the j'th column fine ok. Now that is pretty much the explanation right? So now, the C ij'th entry is supposed to be given by this formula. If the means are 0 you are just left with this formula. And this is nothing but the dot product between the i'th row and the j'th, I mean the i'th column and the j'th column is that fine.

And now if you write it as a matrix then you can just say that it is the ij'th entry of the X transpose X matrix everyone gets this; no one has any confusion the people who raised their hands fine good.

(Refer Slide Time: 20:09)



So, we now this is where the we are so far, that we have assumed that someone has given us these dimensions' p 1 to pn, which we have put in the matrix p right? And we have also made a case that X into P which is what I have written here actually is just a projection of the original data onto this new basis right? Everyone gets that ok. And I am calling that new projection or the new result that I get as X hat. So, that is, what my transform data is.

What is missing here?

Student: (Refer Time: 20:42).

We do not know what p is that I am assuming someone has given me that P, now I need to figure out what is the P here. Now using the previous definition, we get that this is the covariance matrix of the transform data. So, let us just write that this is fine this is fine, what is this?

Student: (Refer Time: 21:03).

Covariance matrix of the original data ok; So, I will just write it as sigma fine ok. Now each cell ij of the covariance matrix towards the covariance between columns i and j of X hat, where X hat is the transformed data, what is the property that you want to hold. I give you 2 conditions or I will give you only one condition for now when i is not equal to j.

Student: (Refer Time: 21:28).

0 ok, so, what should the covariance matrix look like?

Student: (Refer Time: 21:37).

Remember that this is, what is this? This is the covariance matrix of the transformed data right that is what I started with right? This is the covariance matrix of this transformed data, what do I want this covariance matrix to look like?

Student: Diagonal matrix.

A diagonal matrix ok because I want every non diagonal element to be 0 right; And this point I am not telling you what I want the diagonal elements to be I am just telling you I do not want them to be 0.

Well if it is 0 what would that mean?

Student: (Refer Time: 22:05).

That is the variance right if you take the along a diagonal, what you get is the variance it is if it is not clear right now well return back to that. Right now we just know that the off diagonal elements are the covariance between the i'th and j'th column and we want that to be 0. So, we want this condition to hold. This is something very new that you have never seen in this course before they have actually not seen in this course before have you seen this or not?

Student: (Refer Time: 22:3).

Thank god fine.

So, what is this?

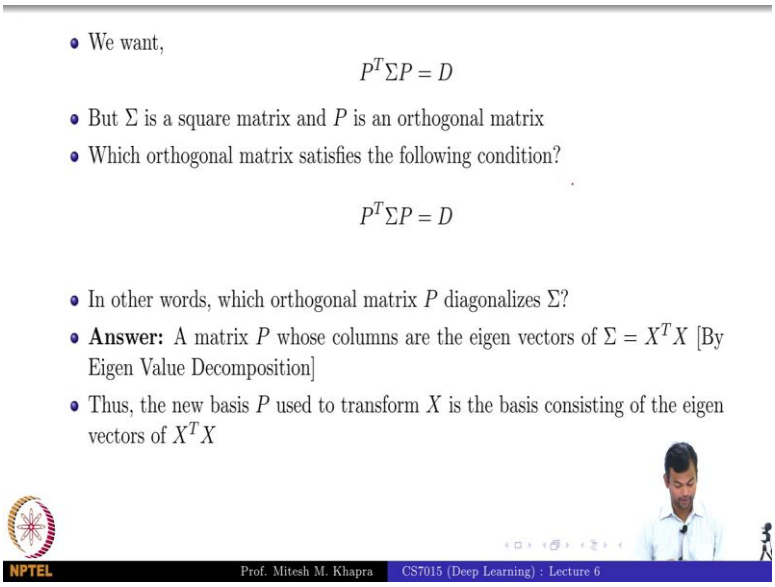
Student: Diagonalization.

The diagonalization of which matrix? This matrix right and what was this matrix it was X transpose X this is clear. So, what is the solution? All rows always lead to.

Student: Eigenvectors.

Eigenvectors, right.

(Refer Slide Time: 22:51)



So, we want P transpose sigma P to be a diagonal matrix and we know which are the set of vectors which I put in; P such that they will diagonalize sigma.

Student: Eigenvectors.

Eigenvectors of.

Student: (Refer Time: 23:06)

X transpose X right ok, wait why did I put this; it is the matrix of the eigenvectors right? So, it is a matrix of the eigenvectors of X transpose X.

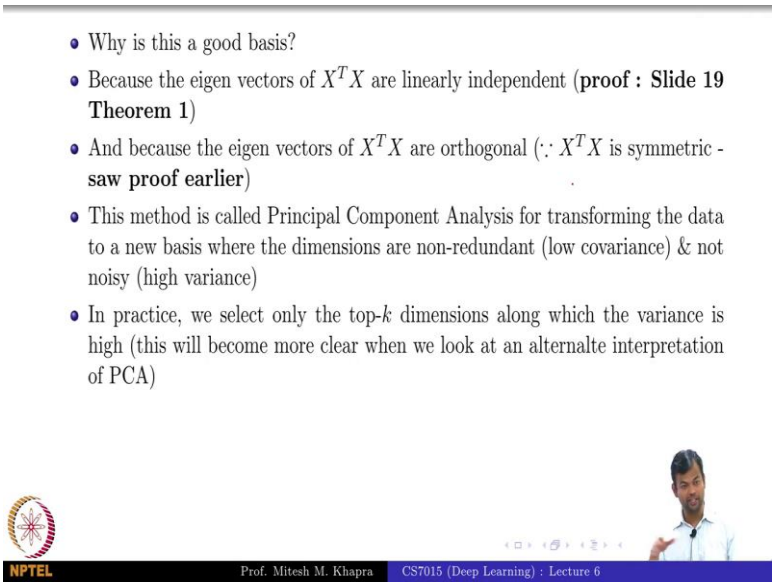
So now have we finished it, do we know principal component analysis now. So, we started with the intuition that we wanted to transform the data ok. I cannot stress enough that we want to transform the data not chopped off dimensions from the existing data ok; that means, we need to project the data to a new basis and we had a couple of conditions the variance should be high, and the covariance should be 0 we have satisfied one condition which is the covariance is 0. And we arrive at a solution which says that the eigenvectors forms the basis that you should project on; so, that the covariance would be 0.

So, we have a solution, we know exactly which basis to use to represent the data ok; so, that the covariance condition is satisfied, what about the variance; did we do anything about the variance?

Student: (Refer Time: 24:10).

So we will come back to that, ok fine.

(Refer Slide Time: 24:08).



Why is this a good basis; what does the what is a good basis, the best basis?

Student: Orthogonal.

Orthogonal right because the eigenvalues of X transpose X are linearly independent that ok and they are also orthogonal because X transpose X is a dash matrix.

Student: (Refer Time: 24:24).

Ok.

Good, real symmetric.

So, this method is called the Principle Component Analysis for transforming a data to a new basis. And that where the dimensions are non-redundant because they have low covariance and not noisy, because they have high variance. The second part I have not

proved right and I will get to that at some point fine. No that is what we saw right no, what is I did not get that now in practice how many eigenvectors would you have.

Student: n eigenvector.

N eigenvectors, do you want to keep all of them? Which ones will you throw away.

Student: (Refer Time: 24:58)

The low variance ok.

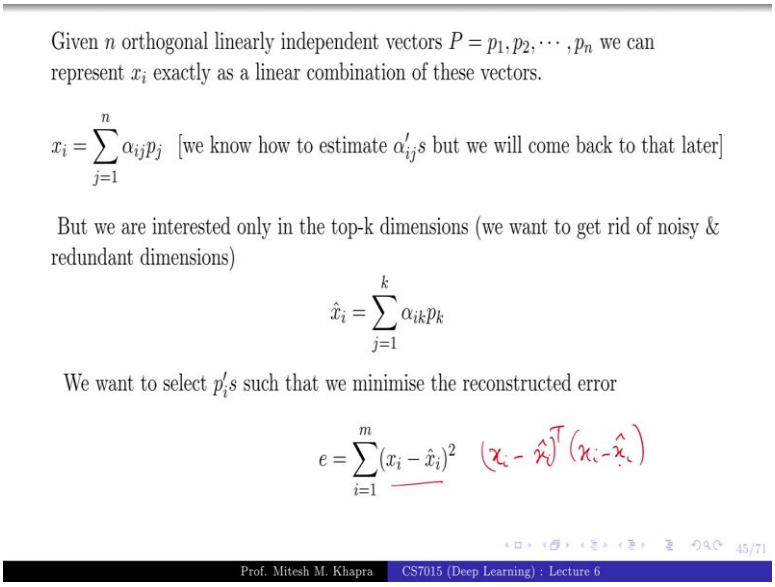
And now in the next interpretation actually we will try to see, what is the; what happens when you throw away the least important dimensions right? What do you mean by the least important dimensions?

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Module - 6.5
Lecture – 06
PCA: Interpretation 2

So, that is what we look at in the second interpretation of PCA right.

(Refer Slide Time: 00:17)



So, again we have the same setup that given n are linearly independent for n orthogonal vectors. We can represent x i exactly as a linear combination of these vectors, what do I mean by exactly? Perfect ok; if you actually describe the whole things in words ok. So, that is exactly what I mean right. So, you are going to write x i as alpha 1 i into p 1 plus alpha 2 i into p 2 and so on. And when you do the summation on the LHS on the RHS, you just get back the LHS when you do the summation on the right hand side you get back the left hand side.

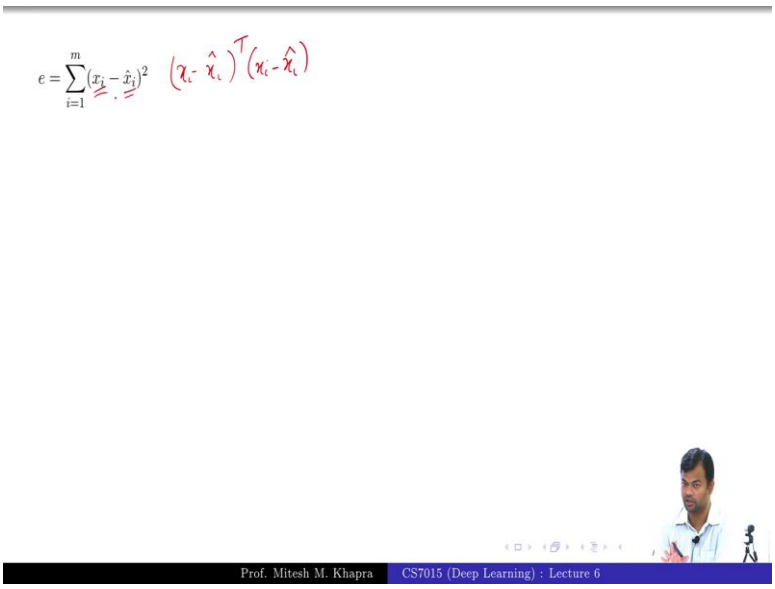
So; that means, it can exactly be represented, when you use all the n eigenvectors now, if I start chopping of stuff what will happen?

Student: (Refer Time: 01:04).

It will just be an approximation ok. Now we this is what I meant, and this is this the equation holds; that means, this is exact and we know how to find the alpha is, because p js are conveniently orthonormal. So, we know how to find that easily ok. Now what if we consider only the top k dimensions, what is going to happen? There is going to be some error in the reconstruction I am not capturing all the information in my original data, but there is some error which I am not being able to capture, and I made a conscious decision that that error is not important I am willing to let it go.

Hence I want to represent the data using fewer dimensions ok. So, this is exactly what you do in PCA when you take the top k dimensions is this fine ok. So now, we want to select pi's such that; we minimize the reconstructed error ok. And this is again erratic actually we should try to write it as, x i minus x, since these are vectors and the square of vectors would just meet this right?

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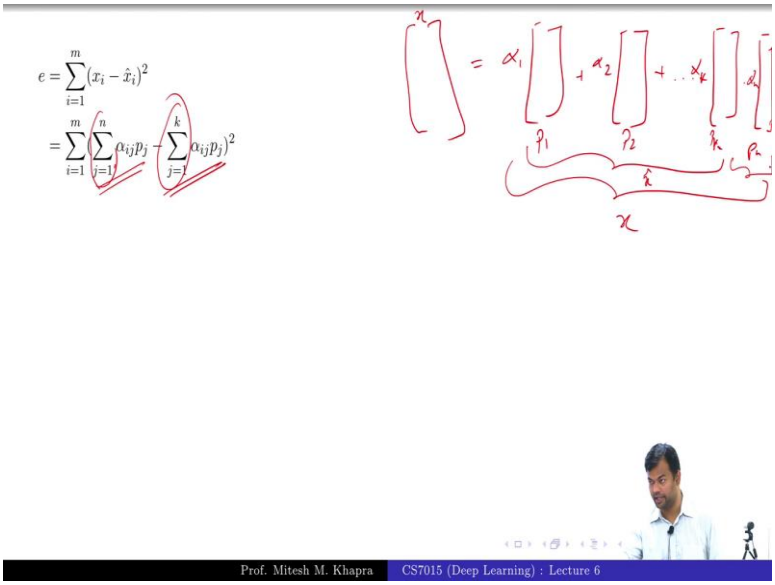


So, but you get the point right were just trying to do the element wise squared error loss were trying to minimize that, we want to do this. So now let us try to see that if you are aiming to do this, what is the condition that, we arrive at ok. So, no I thought I would ask for some changes on this.

For a minute all of you can you just bear with the fact that these are actually vectors and not scalars. So, this square actually does not mean anything it actually means x i minus x

i hat transpose, x i minus x i hat. So, when I use square with vectors this is what I mean is that, everyone can work with that notation fine.

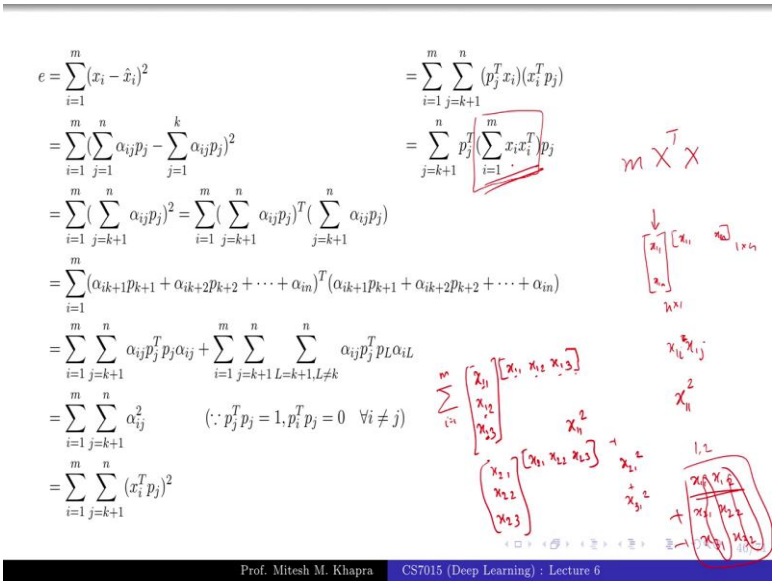
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So now what is x i actually the real point right the correct point which can be obtained by the full reconstruction, if you consider all the n dimensions, what is x i hat just an approximation where you are considering only the k dimensions. Remember that each of these quantities is a vector fine ok. Now what is happening here? Let me just try to say this. So, let me just do this way. So, this is your original x and you are actually writing it as a linear combination of your p's somewhere you will have alpha k pk and then all the way up to p n.

So, this is p k alpha n ok. Now what is this full thing this is x and what is this x hat ok. You see the picture what is the equation trying to tell you ok. Now what is the difference between these two then, these guys right if I want to take difference between x and x hat everyone gets that it is the remaining term say; that means, alpha k plus 1 into p k plus 1 up to alpha n into p n is that clear.

(Refer Slide Time: 04:22)



So, can I write it as yeah can I write it as this ok. So, you get this right. So, I am only taking these guys because the rest will get subtracted. So, one is the full n dimensions the other is only k dimensions. So, if I take the difference between them what remains is k plus 1 to n dimensions, and that is exactly what I have written here ok.

And now I am coming back to the proper notation where this is a vector right. So, I am writing the square as the dot product between the same vector is this ok. These are the m data point right; this sum this is overall the m data points you need to minimize that is that clear ok. So, this is fine, now want just some rearrangement. So, I have just expanded out that summation, this is what it would look like right. I have just expanded out these 2 summations.

Now let us try to do this in your head and see what are the kind of terms that you get there are 2 different types of terms that you will get. So, first of all let us understand that when you expand this you will end up with a lot of dot products, you will get a dot product between this and this and this and so on right. So, can you split those terms into two different types?

Student: (Refer Time: 05:37).

Square terms; So, one where i is equal to j and one where i is not equal to j is that clear fine. So, let me just write it as that. So, I will have k plus 1 to n right; that means, n

minus k terms, where i would be equal to j right so; that means, pk plus 1 was getting multiplied by k pk plus 1, pk plus 2 was getting multiplied by pk plus 2 and so on and then I will have these remaining terms where i is not equal to z right. So, these are the dot product between the other vectors is it fine. You see why I have split it this way, what will happen now? The second term will go to 0 ok; and what about the first term? Alpha ij square ok, now what is alpha ij actually how did you find alpha ij.

Student: (Refer Time: 06:28).

It is a dot product between we did this right finding any of these components is just taking the dot product between x i and that dimension. So, x i transpose pj is that fine ok. Is this fine and again this is slight abuse. So, this is actually, what no this is ok right, a this is ok sorry I am just going to write it as this is this fine. I just written it twice and I can change the order. Since, it is a dot product.

Now, what I am going to do is, so this is actually summation over an index i and a summation over an index j. And I can change the 2 summations I can interchange them ok. So, that is what I am going to do now is this fine. I will push the summation all the way inside what is this actually this entire thing actually m times covariance of.

Student: (Refer Time: 07:25).

So, is this I is this what you are telling me that this is m x transpose x is this fine. How many if you do not get this I see a lot of blank faces how many if you do not get this quite a few; so, this is so i is equal to 1 to m right. So, you are going over the data points ok. So, this what is the dimension of this actually?

Student: n cross 1.

n cross 1 and this is 1 cross n, what does this product give you?

Student: (Refer Time: 07:49).

n cross n what are the entries in this matrix. So, this was say x 1 1 up to x 1 n. And this is again x 1 1 up to x 1 n ok. So, that is going to be x 1 1 square or rather let me just write it in the generic form right. So, it is going to be x 1 i into x 1 j right is that fine. And how many such matrices are you adding?

Student: (Refer Time: 17:07).

m of these. So, what would you get then? What would the first let us. So, ok so, let us do this. So, the first entry of this matrix is going to be x 1 1 square, what about the first entry of the next matrix in this series?

Student: (Refer Time: 08:40).

x 2 1 x 2 1 square right. So, this is slightly tricky to demonstrate, let me just a give me a minute I will just collect my thoughts and do it properly ok. Let us take a small example ok. So, x 1 1 x 1 2 x 1 3 suppose we have a 3 dimensional matrix 3 dimensional data. So, I am taking a sum of m such matrices ok; i equal to 1 to m; that means, this is going to vary this indexes the first index is going to vary from 1 to m. Now, let us see the first matrix and let us look at the first element of that matrix the first element of this matrix is going to be x 1 1 square ok.

Now, let us look at the next matrix, what is the next matrix going to be? It would be x 2 1 x 2 2 x 2 3 right and multiplied by x 2 1 x 2 2 x 2 3, what is the first element of this matrix going to be?

Student: (Refer Time: 09:45).

x 2 1 square what about the third one? x 3 1 square this is fine so far now you are adding all these matrices. So, what is the first element of the resultant matrix going to be x 1 1 square plus x 2 1 square plus x 3 1 square, what is this actually? This is the dot product of x 1 with itself right, and what does that give you the variance if the data is 0 mean right ok. Now can you make a similar argument of the ij'th entry is going to give you the covariance between the I'th and the j'th entry is that clear right. You could do a similar analysis you can actually work it out after going back, how many of you have found comfortable with this? There is still many who are not ok.

So, let us look at an ij'th entry right. So, can someone help me with say that 1 comma 2 entry ok or the first matrix what is it going to be x 1 1 into x 1 2 right for the second matrix.

Student: (Refer Time: 10:52).

x no this is some yeah correct and for the third matrix 3 2 ok. Now what is this sorry what is the summation of these? When you take the full sum you will get these 3 as as, what is this in this summation tell you?

Student: (Refer Time: 11:10),

Covariance between.

Student: First and second

The first column and the second column is that clear now, is it with everyone now,ok fine. So, what you have here is actually the covariance matrix you seems to be lost is it with you sure ok fine.

(Refer Slide Time: 11:32)

$$e = \sum_{i=1}^{m}(x_i - \hat{x}_i)^2$$

$$= \sum_{i=1}^{m}(\sum_{j=1}^{n}\alpha_{ij}p_j - \sum_{j=1}^{k}\alpha_{ij}p_j)^2$$

$$= \sum_{i=1}^{m}(\sum_{j=k+1}^{n}\alpha_{ij}p_j)^2 = \sum_{i=1}^{m}(\sum_{j=k+1}^{n}\alpha_{ij}p_j)^T(\sum_{j=k+1}^{n}\alpha_{ij}p_j)$$

$$= \sum_{i=1}^{m}(\alpha_{ik+1}p_{k+1} + \alpha_{ik+2}p_{k+2} + \cdots + \alpha_{in})^T(\alpha_{ik+1}p_{k+1} + \alpha_{ik+2}p_{k+2} + \cdots + \alpha_{in})$$

$$= \sum_{i=1}^{m}\sum_{j=k+1}^{n}\alpha_{ij}p_j^T p_j\alpha_{ij} + \sum_{i=1}^{m}\sum_{j=k+1}^{n}\sum_{L=k+1, L\neq k}^{n}\alpha_{ij}p_j^T p_L\alpha_{iL}$$

$$= \sum_{i=1}^{m}\sum_{j=k+1}^{n}\alpha_{ij}^2 \qquad (\because p_j^T p_j = 1, p_i^T p_j = 0 \quad \forall i \neq j)$$

$$= \sum_{i=1}^{m}\sum_{j=k+1}^{n}(x_i^T p_j)^2$$

$$= \sum_{i=1}^{m}\sum_{j=k+1}^{n}(p_j^T x_i)(x_i^T p_j)$$

$$= \sum_{j=k+1}^{n}p_j^T(\sum_{i=1}^{m}x_i x_i^T)p_j$$

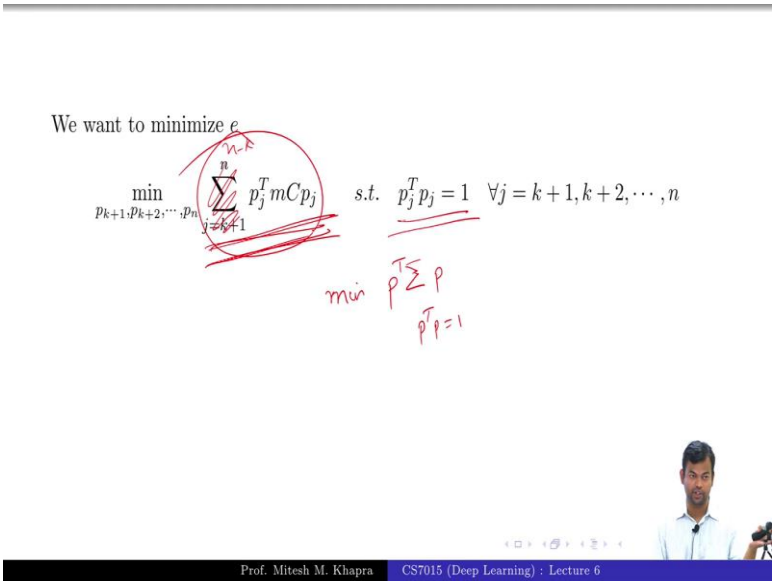
$$= \sum_{j=k+1}^{n}p_j^T mCp_j \qquad [\because \frac{1}{m}\sum_{i=1}^{m}x_i x_i^T = \frac{X^T X}{m} = C]$$

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So, what we have here is something of this form ok.

(Refer Slide Time: 11:37)



So now what we want to do is we want to minimize this quantity subject to the following condition is that ok. What is the solution for this? If I did not have the summation ok; Suppose I just wanted one dimension. So, I want to minimize say p sig p transpose sigma p such that p transpose p is equal to 1, what is the solution for this?

Student: (Refer Time: 12:08).

Smallest eigenvalue of sigma right; and you can show by induction that if you want k such things that here I am looking for n minus k such things right. Then these would be the n minus k smallest eigenvalues of sigma, but now I am talking about the smallest eigenvalues, but in the first solution I said we need to pick the largest eigenvalues. So, what is the difference?

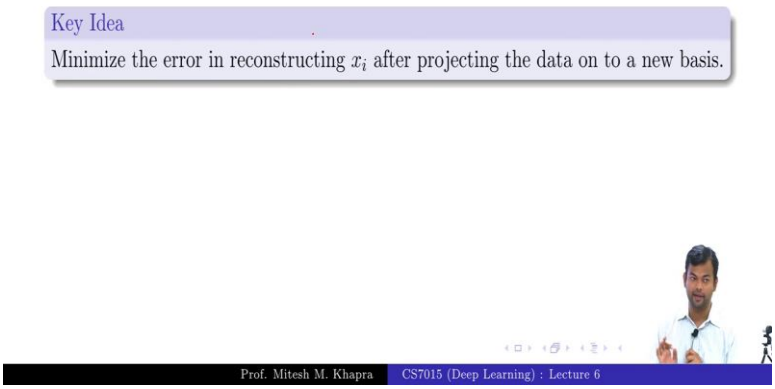
Student: (Refer Time: 12:35).

These are the ones we are throwing away; these are the ones along which the error is going to be minimum if we throw these away the error is going to be minimum. So, we will throw away the last n minus k dimensions which means well keep the first k dimensions is that clear. So, you arrived at the same solution is that right so; that means, in PCA you are actually trying to pick the dimensions in a way such that your reconstruction error is minimized, and this was exactly what our reconstruction error

was. So, do not worry about this math bit, just see that we started with this quantity this is what we wanted to minimize ok.

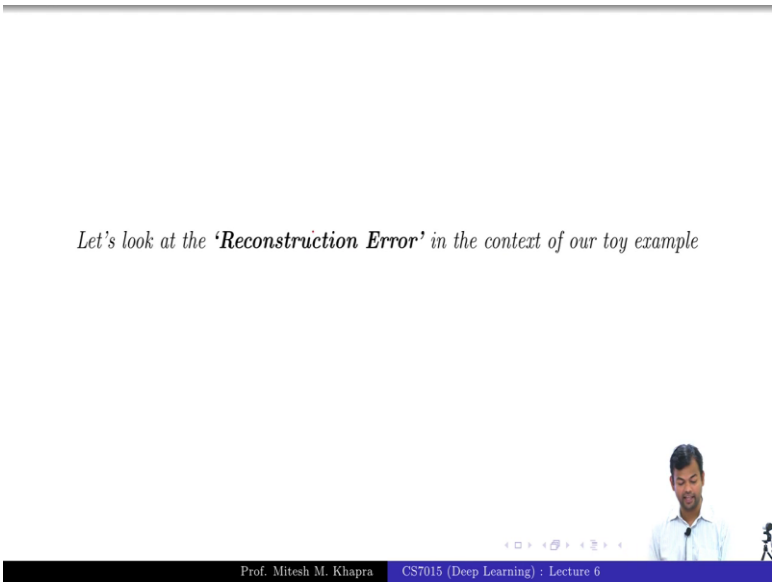
And we did some trickery and we came to this formula that minimizing that error is equivalent to minimizing this quantity. And for this we know the solution that the solution is the smallest eigenvalue and we want n minus k such things. That means, there would be the n minus k smallest eigenvectors is that clear; that means, we are going to keep only the k largest eigenvectors ok; that means, you are going to project your data on to k largest eigenvectors.

(Refer Slide Time: 13:37)



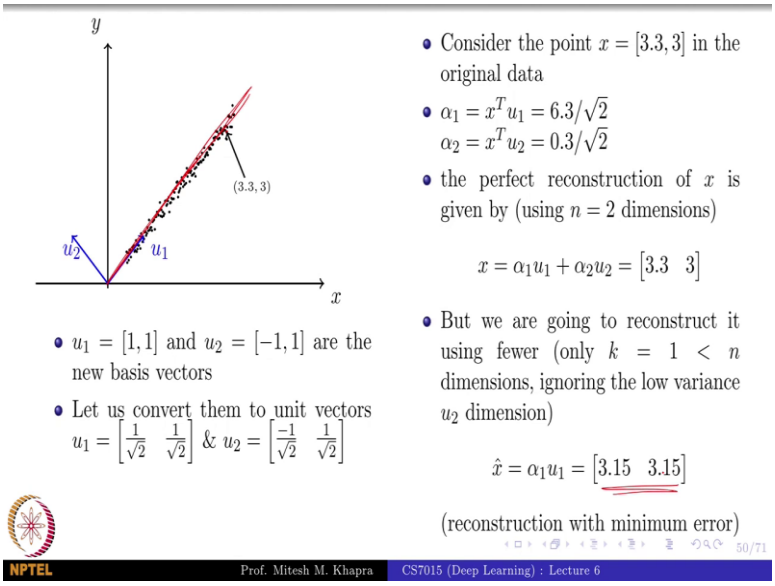
Now, so the key idea here is this right minimize the error in reconstructing x i after projecting the data onto the new basis.

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So, let us take an example and we will work with our toy example again.

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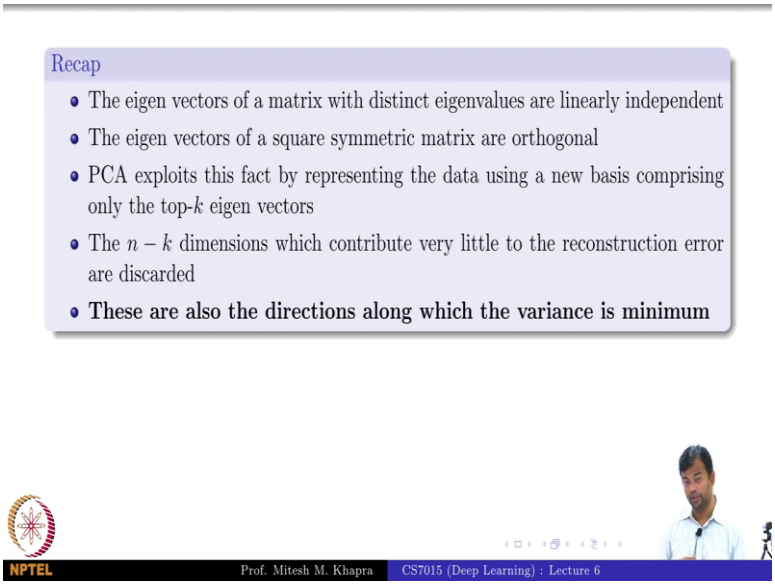
So, this was the data that we had and suppose I give you a new basis which is 1 comma 1 and minus 1 comma 1 ok. This is a new basis this is an orthonormal basis orthogonal basis you can see that u 1 transpose u 2 is equal to 0 ok.

Now, I need convert it to an orthonormal basis. So, I have just divided by the magnitude is it ok fine. Now consider the point 3.3 comma 3, this was our original point according to which coordinate axis x comma x; that means, this was 3.3 and this was 3 ok. Now I

can find the alpha is right because this is an orthonormal basis I can directly find the alpha is, now the perfect reconstruction would be this. So, actually if I do this I get back the original point.

Now, what would happen if I throw away the second dimension, because the second dimension had corresponds to a smaller eigenvalue ok I will get this. So, you see that the point is still close to the original point I have not actually lost much right. What has happened is I have actually projected the boy lie point on this line right, the line x equal to y that is, why I get x equal to y. And in doing that I am not losing much information from the original data is this clear right. So, you understand what happens when you reconstruct the data.

(Refer Slide Time: 15:18)



There is no end to this ok. So, just to recap the eigenvectors of a matrix with distinct eigenvalues are linearly independent. And we use this fact conveniently at least in the case of square matrix where the also happen to be orthogonal. So, we know that they can form a very convenient basis and PCA exploits this to find the top k eigenvectors which to be retained.

And while doing this they have seen that two things are at least ensured one the covariance between the dimensions is 0 because that is exactly how we formulated it and found the solution. We saw that it turns out that we need to diagonalize a certain matrix and the solution is the eigenvectors.

We also saw a different interpretation where we saw that it is the same as throwing away the dimensions along which the error would be minimum right. And both these interpretations led to the same solution which was project the data onto the eigenvectors of the covariance matrix of the original data. ok And this n minus k dimensions current contribute very little to the reconstruction, now what is the one thing which I have not proved yet? What was our wish list?

Student: Variance and covariance.

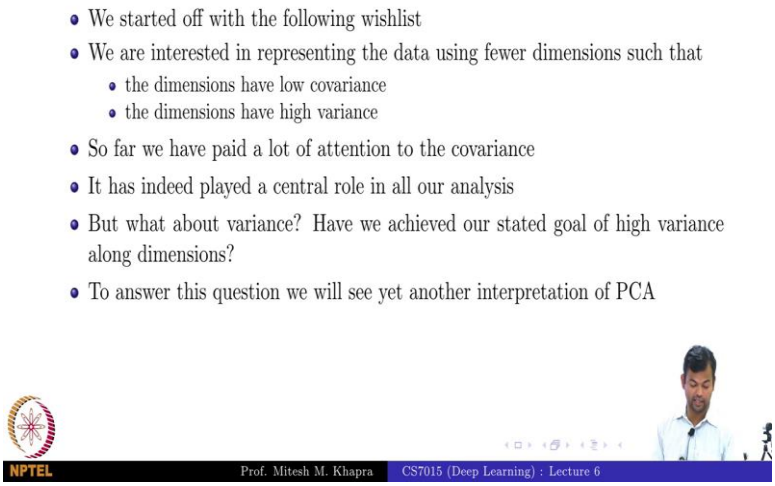
Variance and covariance right high variance low covariance. I proved low covariance, I have also proved something with respect to reconstruction error because that is something I require for auto encoders. So, just remember this bit about reconstruction error.

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Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Module – 6.6
Lecture - 06
PCA: Interpretation 3

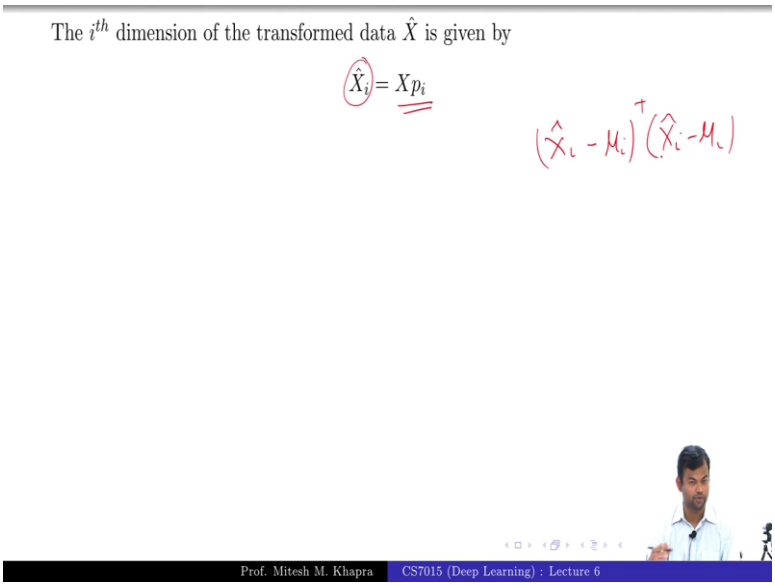
Now, you go to the third interpretation, where we will try to say something about the variance.

(Refer Slide Time: 00:18)



So, we started off with the following the wish list, that we wanted low covariance and we wanted high variance. So, far we have paid attention to the covariance because everything was revolving around this covariance matrix in both the solutions, but what about variance have we achieved the goal with respect to high variance.

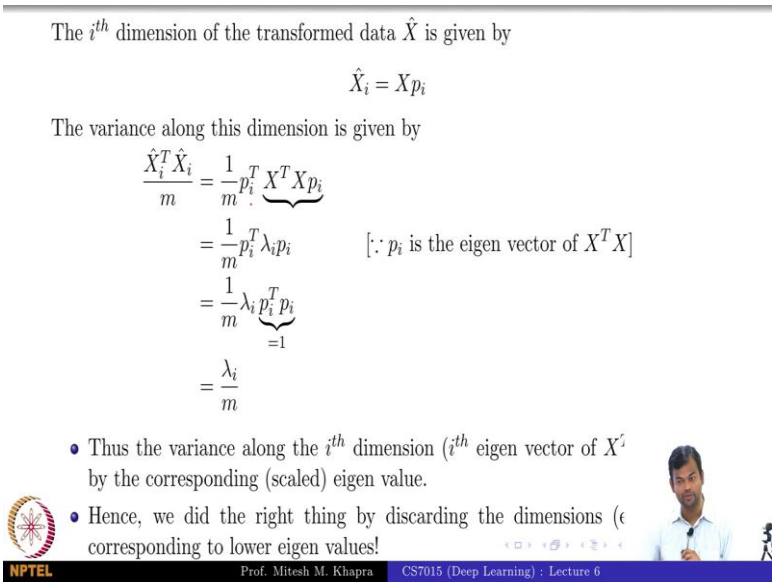
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So, let us see. So, what is the i'th dimension of the transformed data it is this, you take your data and project it onto the i'th dimension right. So, X hat is equal to X into pi now what is the variance along this dimension, how do you compute the variance? So, this is my projected data and let me just call it X hat i.

So, this is the i'th column after projection is that fine everyone is ok with this? Now, for this i'th column I want to compute the variance; how will I do that? Remember that the data is zero mean what is the formula actually? It is going to be X hat i minus mu i into X hat i minus mu i right, but mu i is 0. So, it just turns out to be the dot product dot product of X i hat with itself ok and of course, divided by m is this fine?

(Refer Slide Time: 01:34)



I can write this as X p i and then when I take the transpose I will get this ok. Now what is this quantity? This is exactly the moment where I feel like saying [FL] what is this quantity?

Student: (Refer Time: 02:00).

No look at the circle, what is X transpose x times p i?

Student: (Refer Time: 02:08).

What is p i with respect to x transpose x?

Student: Eigenvector eigenvector

Eigenvector so what is this product going to be?

Student: Lambda (Refer Time: 02:14).

Lambda i p i is that fine, what is p i transpose p i?

Student: 1.

1 ok so what is actually the variance along the i’th dimension?

Student: Lambda (Refer Time: 02:26).

What is lambda i?

Student: Eigenvalue.

So, what will happen if I retain the highest eigenvalues?

Student: (Refer Time: 02:33).

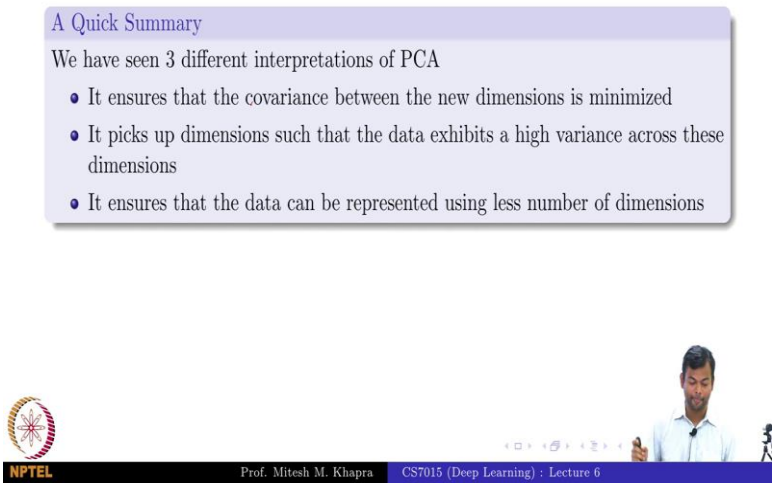
I will get the highest variance dimensions right fine. So, all roads lead to.

Student: (Refer Time: 02:39).

Eigenvectors eigenvalues right. So, Andrew NG in one of his lecture says that there are 10 different interpretations of PCA, I only know 3 of these I do not know the remaining 7. Maybe he was bluffing so that people like us can keep busy oh this is getting recorded.

So yeah, so, you get this. So, we have satisfied everything in our wish list variance, covariance and also did this detour where we saw that it actually amounts to minimizing the error in reconstruction where we are throwing away the dimensions, along which reconstruction did not add much to our knowledge about the data. So, these are the three different interpretations that I have right. So, hence we did the right thing by throwing away those dimensions, which correspond to the lowest eigenvalues because lowest eigenvalues is nothing the lowest variance also.

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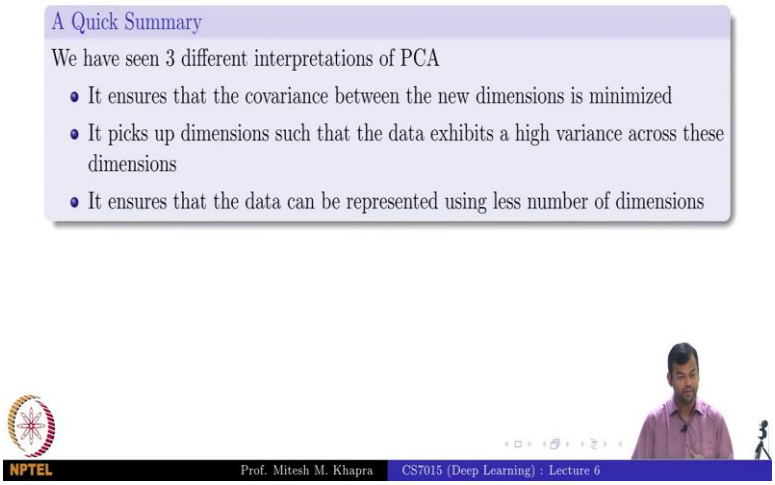


So, this is the quick summary, the covariance between the new dimensions, you can leave actually those you can just read it later on.

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Lecture - 06
PCA: Interpretation 3 (Contd.)

(Refer Slide Time: 00:11)



A quick summary, we have seen 3 different interpretations of PCA, and eigenvectors played a very crucial role in that; and the other thing which played a crucial role was the covariance matrix of the original data. And with these three different interpretations what we realize is that, the solution that we get or the transform data that we get projecting the original data on to the on to a basis consisting of eigenvectors, ensures that there is high variance across the new dimensions.

And we can ignore of the bottom top n sorry bottom n minus k dimensions along with these variance is not high. This also ensures that the error in reconstructing the data by ignoring this dimensions is minimized right it is a lowest possible error. And it also ensures that the covariance between your retained dimensions is 0, because we are able to diagonalize the covariance matrix of the transformed data so that is what we had.

So, now if you think of it right just to connect it two things that we need later on for auto encoder right? We are trying to learn a new representation for the data right and we are

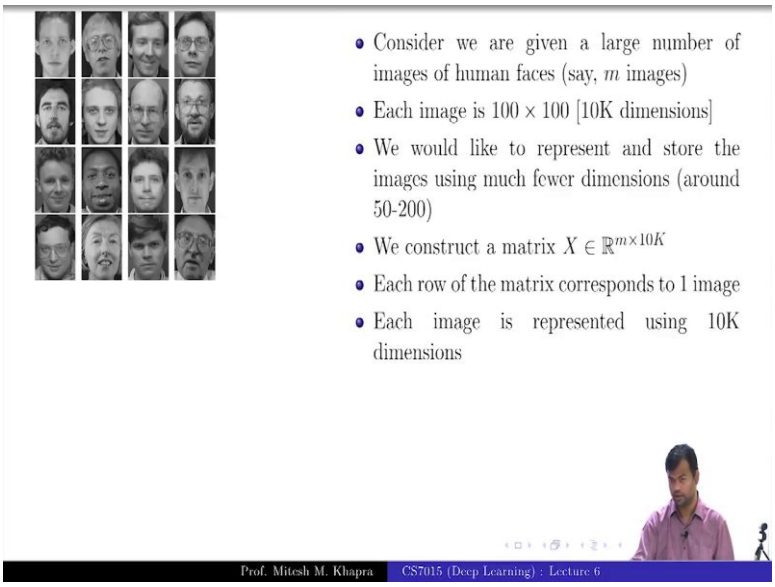
trying to also compress the data, and we want this compression to be such that it is as lossless as possible right? We are going from n dimensions to k dimensions, and still we want to retain the essence of the data and do not want to lose out much of the information in the data ok. So, that is essentially what PCA is doing. Now let us see this in practice.

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Module - 6.7
PCA: Practical Example
Lecture – 06

So, we will in this module we will look at practical example where PCA is used and I just like to give you a flavor of why all this is important right; why do we need to throw away some dimensions and then how does it practically help.

(Refer Slide Time: 00:26)



So, consider that we are given a large number of human images right. So, this is like some faces data set; a database that says one of the intelligent agency someone is maintaining, one of the government agency or may be Aadhar data bases or something like that ok. Now each image here is 100 cross 100; that means, it is 10 K dimensions right? It is a very high dimensional data ok. And your job is to actually store this on to do some database for a large amount of the population right because you are collecting these images from various people.

So, now we would like to represent and store this data using much fewer dimensions right? And you would be really ambitious that if you want to store that more than 50 to 200 dimensions right? So you see the compression that I am looking at. You have 10 K

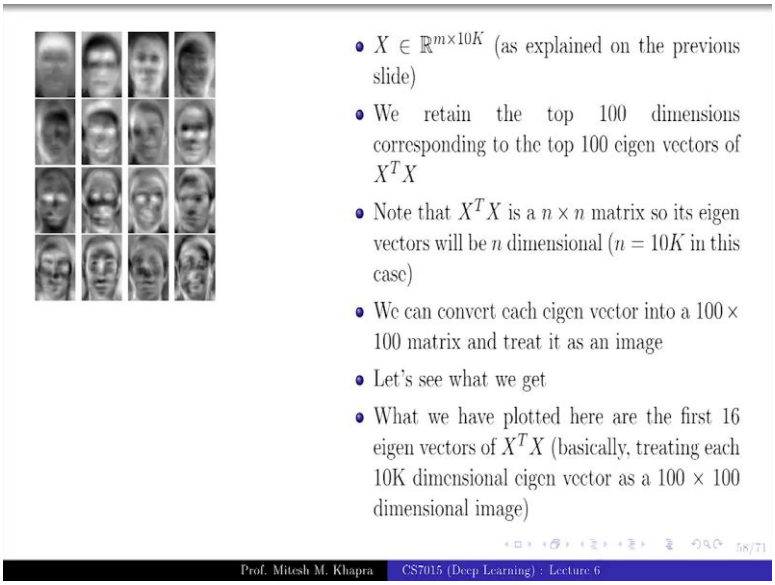
which is a big storage problem for me and I want to just bringing out to 50 to 200, but I have know that this is crucial data right? I do not want to store information which is not able to distinguish these faces, I was still be able to reconstruct the faces from this information right? Do well I mean minimum error reconstruction from this and that is exactly what PCAs are allowing us to do right. So, now we construct a matrix of m cross 10 K; what is m?

Student: (Refer Time: 01:46).

The numbers of samples you have, the numbers of data points that we have and each of this is of dimensions 10 K. Ok, so, this is what matrix, what do we call this matrix? Oh it is already given right; it is the X matrix the data matrix that we always have. Now each row of the matrix corresponds to one image and each image is represented using 10 K dimensions just to reiterate.

Now, let us see; so now what would you do? This is the original data, I want a dimensionally reduced data right? You want store this ah; is the mike working? You want this data to be represented by a fewer dimensions. So, what is your solution? Do PCA. So, what will you do? X transpose X right and I did not get my slide (Refer Time: 02:43).

(Refer Slide Time: 02:44)



So, we retain the top 100 dimensions corresponding to the top 100 eigenvectors of X transpose X right? So, basically we do a PCA, find the 100 find all the eigenvectors of X transpose X and then just retain the top 100 of those. Now what is the dimension of each of these eigenvectors? Should be straight forward take your time, it is early morning.

Student: (Refer Time: 03:10).

10 K right. So, now can you think of a physical interpretation of this? So, what are you trying to do? You are trying to store faces and now you have come up with these dimensions; no sorry we have come up with these basis vector which is eigen vectors and each of them is also 10 K; which is as same as dimensions of your faces.

Can you think of a physical interpretation of what is happening here? None of went you through the slide except perhaps you or I do not know just think about it. So, what you are trying to do is you are trying to represent any possible face in your database, right using a linear combination of some vectors ok. Now these vector should have some interpretations right? It should be connected to faces and somewhere; otherwise how will you construct a face from taking a linear combination or some random vectors; do you get the point?

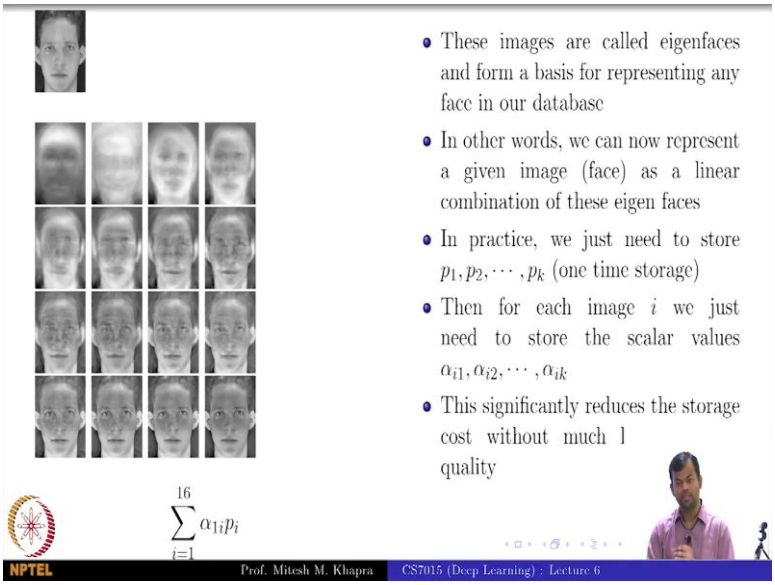
So, can you think of each of these 10 K dimensional vectors which is the same as the dimension of your original data; as a face and try to plot it? Can you try to do that at least it make sense 10 K dimensional ok, that is the same what you are image size was. I could just arrange these 10 K dimension as 100 cross 100 and try to plot it ok. So, let us see if you do that what happens ok; we convert each eigen vector into 100 cross 100 matrix and treat it as an image and let us see what we get.

This is what we get. So, this is the top 16 eigen vectors that I have plotted; now can you tell me a physical interpretation of this? This is the basis for constructing any face in your data base right? That what you are trying to say all the faces that you have in your database or in the world you can combine them by looking at the these elementary face structures right, which are your basis.

And then you could scale them up by using these alphas, you will be multiply them with the certain alpha right? And when you combine them you will get the base any face that

you had in your original database. Does a physical interpretation make sense, how many of you get this?

(Refer Slide Time: 05:28)



So, that is what is happening here. So, we have constructed this basis now I will come to that later. So, these images are actually called eigen faces and the form of basis for representing any face in our database ok. In other words, we can now represent a given image as a linear combination of these eigen faces. So, this is my original image ok; I want to reconstruct it. Now use 16 or 25 of these eigen faces; what do you think would happen? We will get some face which has some error; there is some error in reconstructing this face.

So, let us see what we get. So, I am using only one basis vector and I found out this alpha one I how would I found it out?

Student: (Refer Time: 06:11).

Dot product of the face vector with the basis vector ok, now I instead of 1; I take 2. You see I have took this 2 basis vectors, scaled them with the corresponding alpha coordinates and added it them up right? And I am trying to get some face value it does not look it goes to the original face; that means, the dash is very high; the reconstruction error is very high.

That means, I have still dropped some of the important dimensions; I have still drops some of the important eigen vectors right? So, the value of K which is the top K eigen vectors is something that I need to take care of it and should be in a way. I can always construct the reconstruct; I can always compute the reconstruction error here right? How will you compute the reconstruction error?

Student: (Refer Time: 07:00).

Take the square error between the original image and this second image that you see here right? So, you will take the square error between this and this and you will end up with the number which is not acceptable right? Now what I will do is I will go further I have taken 4 ok still not quite there, but I can see a shape emerging right.

I go to 8 things becomes better; since you are already know what the original face at least you can make sense of it. And by the time I reach 16 I am almost there right; at least I can recognize the face that is probably losing out some subtle things in the face, but I can recognize it.

Now, how many of you appreciate what is happening here? Yeah of course, now what is happening here? So, think in terms of a practical application right; what have you done, what have you able to be achieve, how may basis vectors where you able to store or did you store?

Student: 16.

16; that means, 16 into 10 K values ok and suppose you had a million images in your database; how many would you require to store?

Student: (Refer Time: 08:11).

Wait let us we do it step wise forget about PCA, if you had a million image in your data base and each of them have 10 K dimensions how much storage do we need?

Student: (Refer Time: 08:24).

Million into 10 K floating point values ok. Now with if I say 16; 16 may be too ambitious may be later on I will say 50 or 100 is ok, but let us say 60 then how much data we need to store?

Student: 16 into 10 K.

16 into 10 K and you can reconstruct any face?

Student: (Refer Time: 08:49).

Yeah alpha's needs to be stored right? So, for every image instead of storing 10 K dimensions, you will just store the 16 alphas right? And you can see that even if I go there to 100 and its still manageable, instead of 10 K I am going to just store 100 alphas right? And as I go to 100 what would happen?

Student: (Refer Time: 09:09).

The reconstruction error would become even lesser ok. So, is that is the intuition clear ok? So, this eigen vector storage is a onetime storage; we are going to store this K eigen vectors, each of them are 10 K dimensional and K is 100 or 200; we do not really care because the original data was very large right? And for each image we just need to store these alpha values K of them right? So, for each of them instead of 10 K, we will store 100 to 200 alpha values and of course, it is significantly reduces right.

So, this is why we need to do all this right and what is the other advantage of doing this? Anything of; something else; so what is PCA actually allowing you to do? If you again think of it not; I mean subtract the math out, just think it in terms of physically interpretation and what is that it is allowing you to do? If you had to say it in English what is it allowing you to do? Compression is a loaded word can you just spell it out for me what is this compression mean actually?

Student: (Refer Time: 10:14).

Right. So, it is storing all the relevant information in the image and discarding all the error element information right? Now this also ensures that if you have multiple images of the person, then what would happen? You should take the image under lighting conditions or may be at person had applied some makeup or something like that right what would happen?

Student: (Refer Time: 10:38).

In the original space, the 10 K dimensional space what would happen to these images? They will be very far from each other right because the lighting conditions have changed, you see a dark person, so have a light person something like that right? And now because PCA has helped you to throw away this dimensions right? May be the exact terminology which I am using may be the lighting condition do not do it. Because you can imagine that there would something right that suppose as some element which is calling the image to look slightly different, but that is not the important information right? So, that would get discarded off and only the relevant information would stay.

So, then multiple images of the same person which were dash in the original space would come dash in the new space.

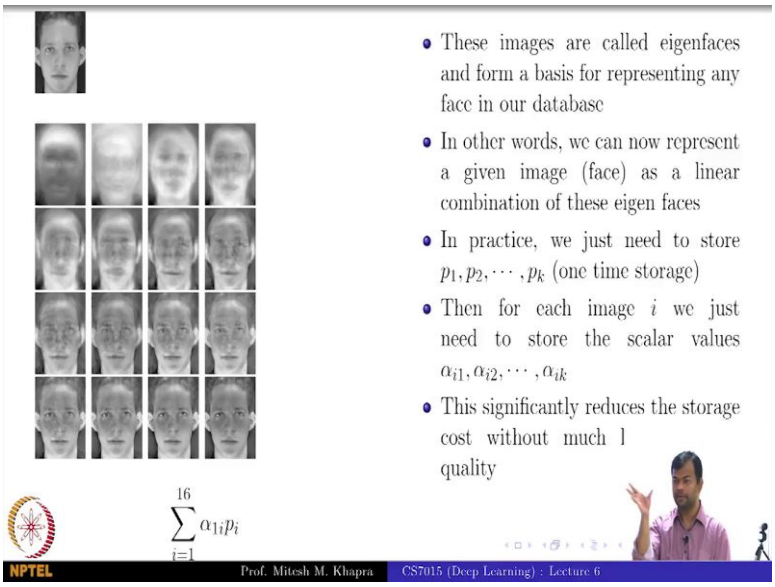
Student: (Refer Time: 11:22).

Far in the original space would come closer in the new space right? So, this is what compression helps you to do; so, this is what you want to learn you want to learn the important characteristics of your original data and that is what PCA allows you to do fine.

Deep Learning
Prof. Mithesh M. Khapra
Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Module – 6.8
Lecture – 06
Singular Value Decomposition

(Refer Slide Time: 00:17)



So, with this we will move on to the last topic in the yeah so, that is something that you will have to so, the way I would do it, right is that you keep aside some 100 images from your data as validation data. Now, once you have learned these eigenvectors, try to compute the reconstruction error for these 100 images. And just vary it, do 100, 1000, 10,000 written as many dimensions as you can, and see at what point is a reconstruction error, ok for you, right and this is assuming that you have some notion of what is a reasonable reconstruction error. So, we all know that the minimum is 0.

But if you have 0.5, then maybe for face database it might be. But if it is a database where you are trying to look at mechanical parts? So, suppose you are looking at motors and rotors from a machine assembly. Now there you want to be able to distinguish minor detects defects on this and a detect could a defect could actually just be one single or 2 pixels getting different from the original image, right? So, there the reconstruction loss would be much needs to be much more robust, you get the point? So, it depends on your

application. So, you will have to take some validation data either have a domain expert to tell you what is reasonable or go by the number that you get, right? And this is the validation error that I get.

So, everyone understands the question and perhaps the answers, ok. So, we now go to the last module.

Student: (Refer Time: 01:40).

Yeah, if you can.

Student: (Refer Time: 01:41).

Yes, you can now project any face into this database A. So, that is the Eigen basis that you have got, you have got the basis vectors, now any data you can project onto this basis.

Student: (Refer Time: 01:54).

Now, so, if you are trying to learn these eigenvectors by say using 100 images all of which belonging to a particular demographic, say all Caucasian images, right? And now at the runtime you have an Asian image, then you will have; obviously, have some error right, but you have large even of data, say if you have if you are constructing this from million images, then it should generalize that is I mean just as for any machine learning algorithm.

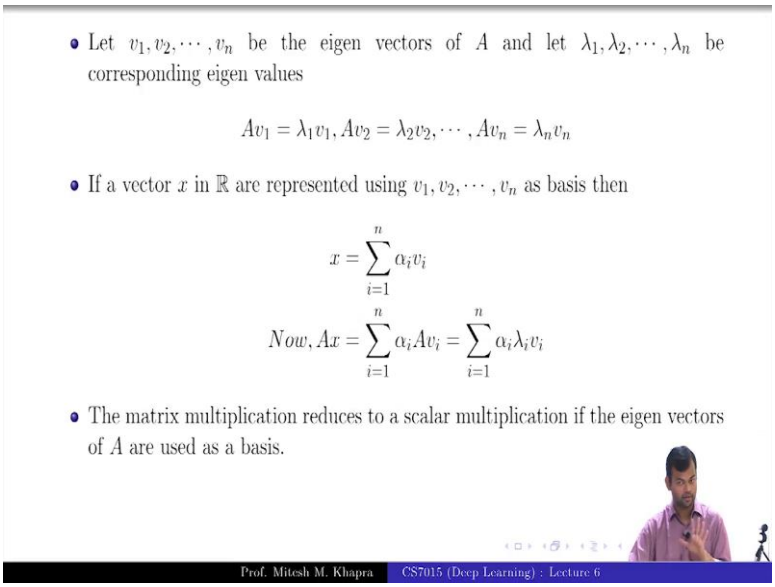
The training it from small data and you bring out some outlier at test time it is not going to work, right? But if you have reasonable data it should generalize, any other questions? To calculate the eigenvectors x is m cross 10 m cross 10 k yes. Now, we move on to the last topic for the basic portion, and the next class we will do auto encoders will be back to Deep neural networks so, singular value decomposition, right?

(Refer Slide Time: 02:50)



So, this is actually the stuff that I need an important theorem from here at multiple 2 places in the course. So now, before doing the, right? Let us get some more perspective on what eigenvectors do and why are they actually important.

(Refer Slide Time: 03:05)



So, let v 1 to v n be the eigenvectors of a and let lambda one be the corresponding eigen values. So, we know this a v 1 equal to lambda v 1 and so on, ok. Now suppose all the vectors in R should be R raised to n, ok. So, if a vector x belonging to Rn can be represented using this basis, ok. Now, what if I am interested in the operation A into x,

what is the advantage of representing it using this basis? So, this is what you are saying the other day?

Student: (Refer Time: 03:54).

What is Ax it is a matrix vector multiplication, right? And it is going to be a heavy computation. Now if all my vectors in Rn are represented using the eigenvector as the basis, what happens to this matrix operation?

Student: (Refer Time: 04:16).

It reduces to.

Student: (Refer Time: 04:21).

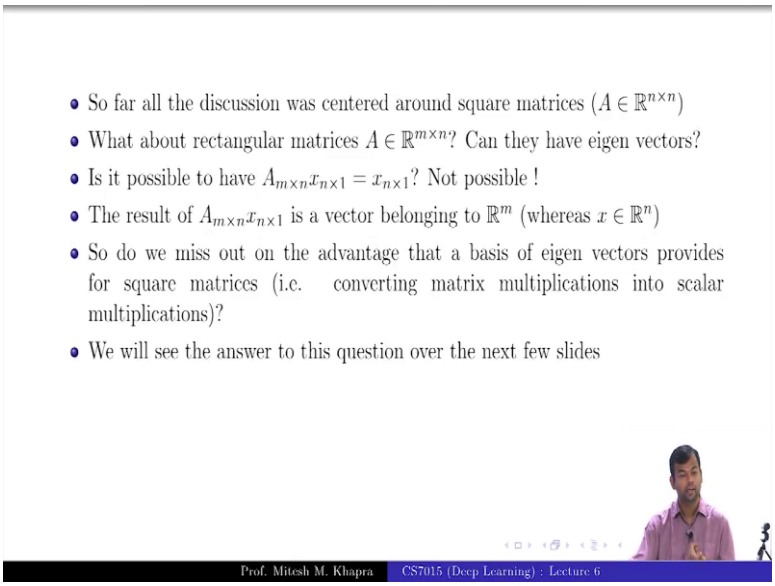
Let us see so, I was interested in Ax, but I know x is this. So, you get this step, and what happens finally? Do you have the matrix anywhere here. So, what happens to the matrix operation?

Student: (Refer Time: 04:41).

It reduces to a sum of scalar operations, right? If your vectors were representing using the eigenvector as a basis.

So, this is one reason why this is important, right? So, you can now get away of the get rid of the matrix operations and just do scalar operations, right? So now, there is a catch here which I am going to ignore, just to try it if I bring in the catch you guys will get confused. So, I will ignore if anyone has a doubt maybe talk to me after the class, but for now let us go with the fact that the matrix operation reduces to a scalar operation.

(Refer Slide Time: 05:19)



Now, so far what we have done is discussed square matrices, I have said that they are the villains of linear algebra, but who are the super villains of linear algebra , rectangular matrices everyone says that, but why. Imagine what they do to a vector yeah. So, can rectangular matrices have an eigenvector?

Student: (Refer Time: 05:45).

Yes; obviously, yes that I mean any matrix can have an eigenvector.

Student: No.

No why? Can you write something of this form? You can't, right? Because when the matrix operates on an n dimensional vector what does it give you?

Student: (Refer Time: 06:01).

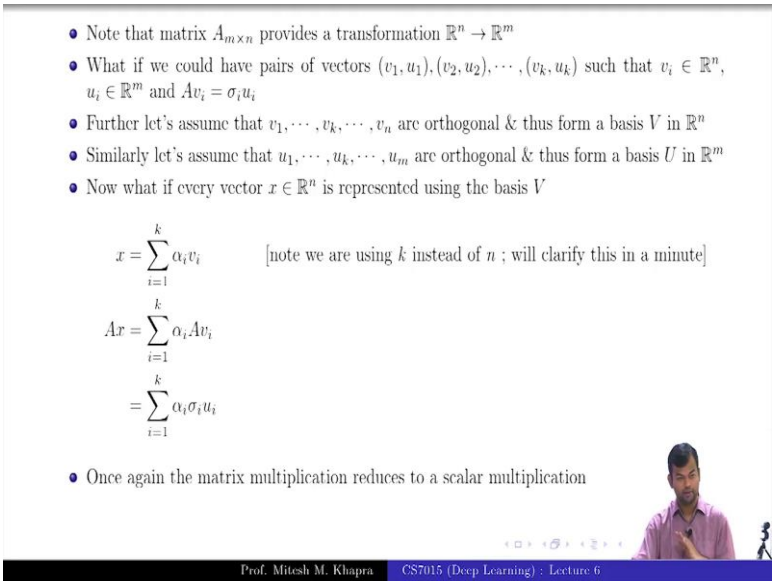
An m dimensional vector, right; hence they are super villain, right? Because they take the vector from one space and transform, it to a completely different space that completely lots lost it is identity, right? So, that is why rectangular matrices are even harder.

So now we just saw that for square matrices this eigenvectors form a very convenient basis where these operations reduce to a scalar operation. But now rectangular matrices do not even have eigenvectors. So, then cannot we have the same advantage there? Can

we have the same advantage there? You can't, right because you do not have an eigenvector, but I would teach you about singular value decomposition. So, I better have something; so, get the connection, ok. There is a problem with square matrices with the rectangular matrices. So now, let us see. So, we will try the aim is to see if we have something equivalent to this scalar transformation that we had for square matrices.

How many of you have seen this in linear algebra before? So, you know whatever I am going to talk about, fine. So, the result of Ax is a vector belonging to R m and the original x belongs to R m. So, we do miss it miss out on this advantage that you could have reduced the matrix operation to a scalar operation, and now we will try to see if we can still get back that advantage.

(Refer Slide Time: 07:32)



So, notice this is matrix, you can think of it as a function; which provides a transformation from Rn to Rm. So, what is the set of inputs to the matrix? It is vectors belonging to Rn . That is the set of input.

Now, suppose we had a pair of vectors v 1 u 1 v 2 u 2 vk u k, each belonging to these 2 different universes one is Rn the other is Rm,. And there was a specular relation between them that A into vi is equal to sigma into ui, suppose I am just being ambitious let us see whether we can actually have this pair, but suppose we had this pair, then can you connect this back to the discussion on scalar operations. So, let us just see that in detail, and we will of course, assume that these are orthogonal and form a basis. So, the vi's

form a basis in Rn and the ui's form a basis in Rm. Is that clear that is all straightforward, we have these vectors.

Now, every a vector belonging to Rn which was the input space can be represented using a linear combination of v straightforward, and any vector belonging to the output space can be represented of.

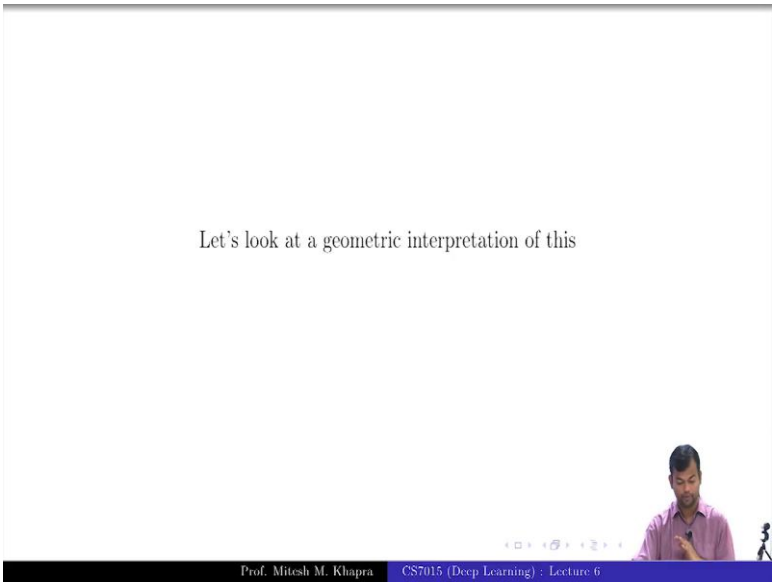
Student: (Refer Time: 08:55).

Of u, right, so, that means, any x in the input space I can write it as this linear combination. And now if I do the matrix operation what happens?

Student: (Refer Time: 09:09).

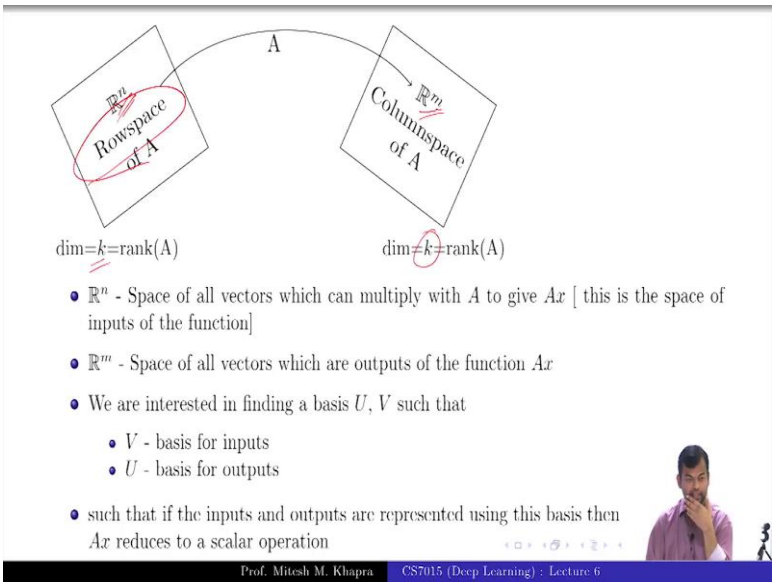
You get this A into vi, what is A into vi? sigma ui, I have still not shown you how to find these sigma is ui by the way, right, ok? Once again the matrix multiplication reduces to a scalar multiplication.

(Refer Slide Time: 09:30)



So now let us try to look at a geometric interpretation of this.

(Refer Slide Time: 09:32)



So, what you have is this original space which is Rn you are using a as a matrix operation, right as a function and you are transforming vectors from n to Rm right? So, this is the space transfer that I was saying it vectors are being picked up from Rn and being put into R m, ok. And Rn is a space of all vectors which can act as inputs to this function, and Rm is a space of all vectors which are the outputs of this function Ax.

Now, we are interested in finding a basis u v such that v is the basis for the inputs, when I say basis all of you should immediately start thinking of dash vectors.

Student: Orthonormal vectors.

Orthonormal vectors orthogonal or orthonormal, right? Once we have orthogonal we do not care about the rest, u is the basis for the outputs; such that if the inputs are and outputs are represented using this basis, then all our matrix operations reduce to scalar operations. So, we are just trying to find the rectangular analogy for the square a phenomenon that we observed, ok. That is what we are trying to do, now can you tell me, I have told you that if such a v and u exists, then you could do this. Can you give me such a u and v?

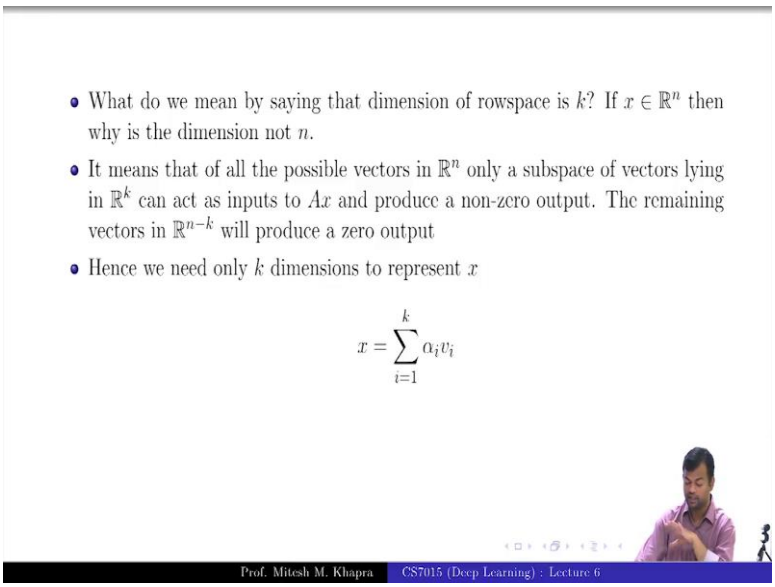
So, what do we mean by so, here I said actually I said this, right, that the dimension of the row space is actually k and the dimension of the column space is also k what do you

mean by the dimension is I mean, right? Here I am telling this is Rn and this is Rm, and now I am telling you the dimension is k, what do I mean by that?

Student: (Refer Time: 11:05).

The only k linearly independent vectors.

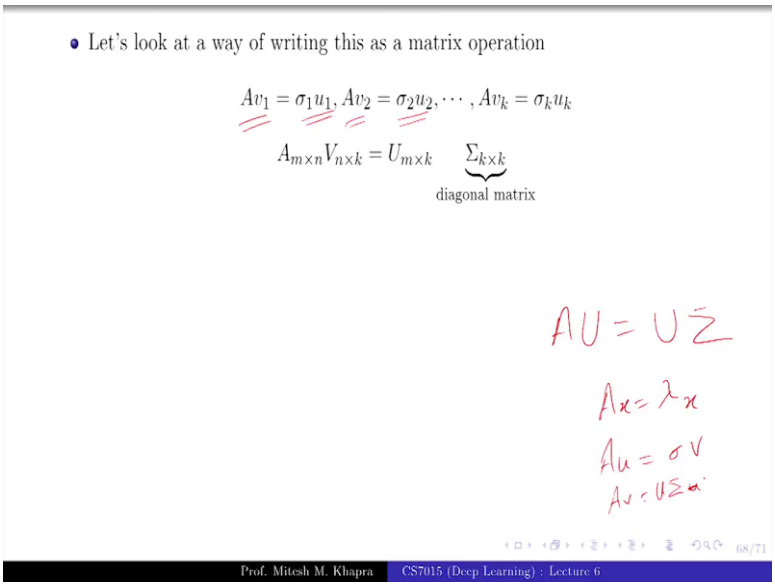
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And this is again something from linear algebra which I expect you to know is that all possible vectors in Rn only a subspace belonging to Rk can actually act as input to a x to produce a non-zero output. So, I am talking about a null space column space and things like that right? So, this should be clear if it is not it is. It is not very important at for us right now.

And hence we have only k dimensions.

(Refer Slide Time: 11:38)



So, let us look at a different way of writing this. So, you have this a v 1 is equal to sigma 1 u 1 Av 2 is equal to sigma 2 u 2. So, I can again do the same trick that I put all the v's into one matrix; where vi's are the columns of this matrix. And I will put all the us into another matrix where ui's are the column of this matrix, is that fine everyone, ok? So far and then I can write it as this matrix operation. Same thing that we did when we are doing eigenvalue decomposition, right? So, we had written it as A into u is equal to u into sigma, right? Because there we had the condition that Ax is equal to lambda x, now we have a u is equal to sigma v or rather the other way around. So, Av is equal to did I missed up, did I no, right?

Student: (Refer Time: 12:29).

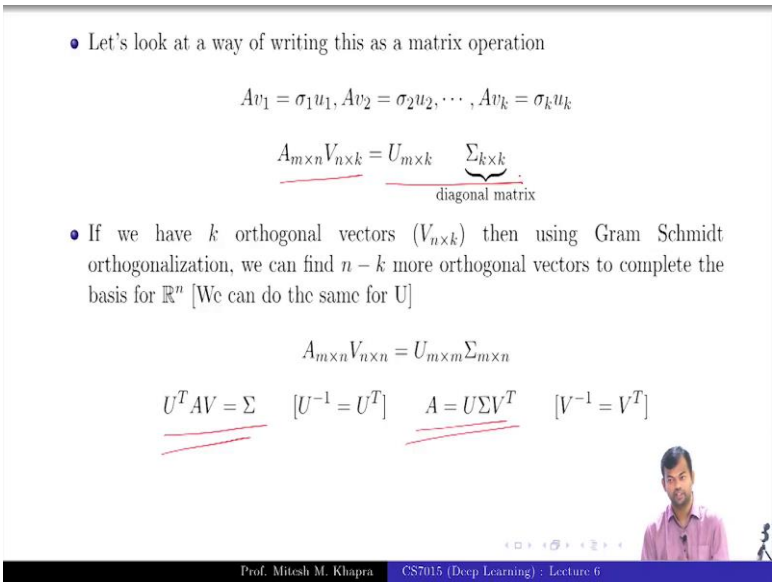
Sorry.

Student: (Refer Time: 12:32).

Fine, yeah so, Av is equal to sigma into u.

So, is this fine no, but when you do the diagonal operation you will get it as u into sigma y. The same way as a x equal to lambda x, but when you write it is A into u is equal to lambda comes later on.

(Refer Slide Time: 12:56)



And if we have k orthogonal vectors; so, remember I said that this basis consists only of k dimensions, right? Because that is R the set of vectors which can act as input to A. So, what I, but I want a basis for the full Rn. So, what do I do for the remaining n minus k? Have you heard this Gram Schmidt Othogonalization, right? So, if I give you if there if you are trying to construct a basis for n, ok. For Rn rather and if I give you k orthogonal vectors, they can do k you can construct the remaining n minus k using Gram Schmidt orthogonalization right? So, you can get the full basis ok fine. So, let me just see and this is orthogonal ok. So, you can write so, you see these 2 forms can you relate it to something that we have seen before in the course.

This is singular value decomposition, what else did we see before?

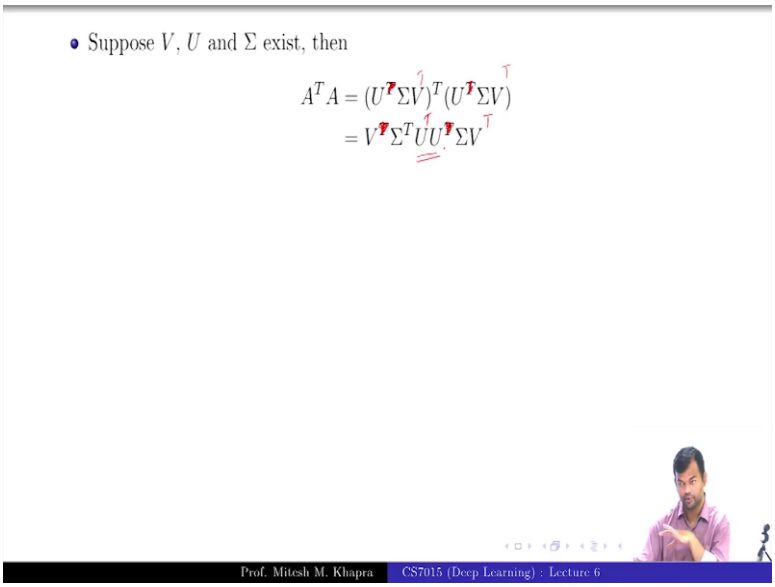
Student: (Refer Time: 13:53).

Eigenvalue, so, this exactly the same forms, right; And I have used the same set of tricks to arrive at it right? So, I first put the vectors into a column as columns into a matrix then wrote this in the matrix format, and then pre multiplied post multiplied by certain things and I got these 2 formats. And remember that v and u both are dash matrices.

Student: (Refer Time: 14:13).

Orthogonal matrices, right? So, that inverse is just their transpose, ok, ok. So far everything is fine, now I still do not know what U and V are, all this analysis is assuming that I know what U and V are. So now can you tell me how to get these U's and V's.

(Refer Slide Time: 14:30)



Suppose v u and sigma exist; then we can write this right? So, A is u sigma v so, A transpose would be the transpose of that. Now can you work with me, what is the next step?

Student: (Refer Time: 14:47).

Ok next.

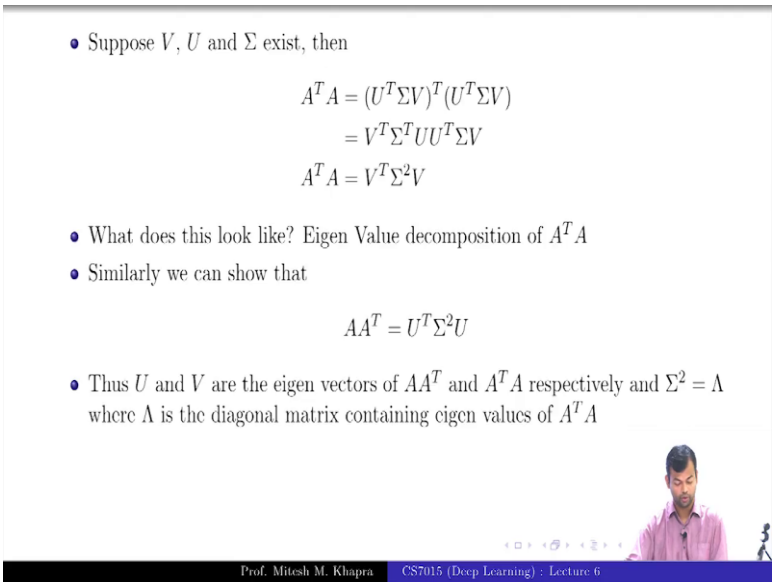
Student: (Refer Time: 14:51).

This is u sigma v transpose. So, then this would be I think the next step is no the next step is also wrong, that fine? Ok fine, I just had some error with the transpose, ok. What will happen now? What will disappear from here?

Student: (Refer Time: 15:26).

U transpose U that is I, right? U transpose the inverse of U.

(Refer Slide Time: 15:30)



So, you get this, what does this look like? This looks like the eigenvalue decomposition of.

Student: A transpose A.

A transpose A; that means, v consists of the.

Student: Eigenvectors.

Eigenvectors of the.

Student: A transpose A.

A transpose A; So now, can you tell me what u would?

Student: (Refer Time: 15:54).

Ok fine.

So, this looks like the eigenvalue of eigenvalue decomposition of A transpose A. Similarly, we can show that A A transpose is equal to u transpose sigma square u, ok. So, then u is the set of Eigen vectors of A A transpose, right? And now here what was with will the eigenvalue decomposition always exist for a matrix.

Student: No.

No under what conditions would it exist? First of all it has to be a square matrix.

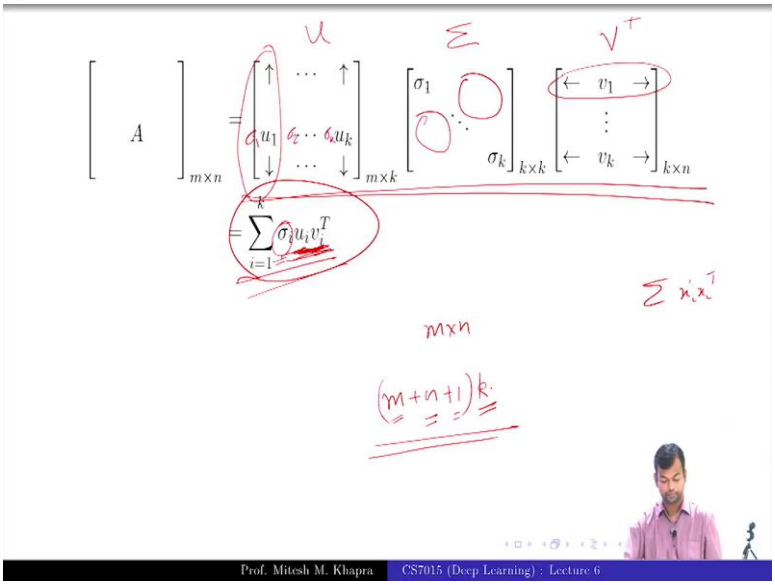
Student: (Refer Time: 16:32).

Ok right, but now for a rectangular matrix would be singular value decomposition always exist.

Student: Yes.

Yes, right? Because it depends on the eigenvalue decomposition of square symmetric matrices, ok. Is that fine, ok? So, for any matrix shall always have the Eigen value oh sorry the singular value decomposition.

(Refer Slide Time: 16:58)



So, this is perhaps yeah, ok, now just one last bit and let us see if all of you can understand this. So now, I can write A in this form this is nothing but what I already said, right? This is u this is sigma this is v transpose, ok. Now from here from this step do you see how I got to this step? This is something that we were struggling with yesterday also, when we were trying to find out summation x i x i transpose something similar here, you know the 4 ways of multiplying matrices, right? So, this is which way one of the ways.

Yeah so, does everyone get this, right? So, a simple thing would be first to just take these sigmas inside, right? Because this is a diagonal matrix, right this is all 0's. So, these are actually you can write it as sigma 1 u 1 sigma 2 u 2 and sigma k u k, right.

Now, this ends up being the product of 2 matrices, right? And you can write it as a sum of columns into rows right? So, what I am writing it as a sum of sigma 1 u 1 multiplied by v 1. So, sigma 1 u 1 into v 1 transpose is a scalar matrix vector matrix right? So, each of these terms here is a?

Student: Matrix.

Matrix, and you are adding k such matrices, ok. Now try to relate it to reconstruction error. You are taking a matrix trying to write it as sum of many matrices. If I trim some terms from this some terms from this sum what would happen? If I have all the terms, then what would happen? I will get A back exactly. If I drop some terms what would happen?

Student: (Refer Time: 18:57)

I get an approximation of A, how good would that approximation be?

Student: (Refer Time: 19:03).

First is depending on the number of dimensions, but is there a natural ordering in these dimensions if I want to throw away some dimensions which one would I throw away.

Student: (Refer Time: 19:11).

Smallest.

Student: Singular values.

Singular values, sigmas are the singular values. So, you see that this is getting multiplied here, every matrix is getting multiplied by the singular value corresponding singular value. So, if I drop out the terms which have the smallest singular values, then those matrices the elements would have been very small. So, I will not lose much in the approximation. So, again the same idea that I am trying to approximate the original matrix by a smaller rank.

By of so now, the original matrix had m cross n entries, ok. How much if I use only k eigenvectors or the sorry k singular vectors or k dimensions to approximate it, how much

storage would I need? How many values do I need? So, the original matrix was m cross n. How many entries are there here?

Student: (Refer Time: 20:06).

Each of this is how much?

Student: (Refer Time: 20:10).

M for ui plus n for vi, ok. And plus 1 for the sigma, and how many of these are there?

Student: k.

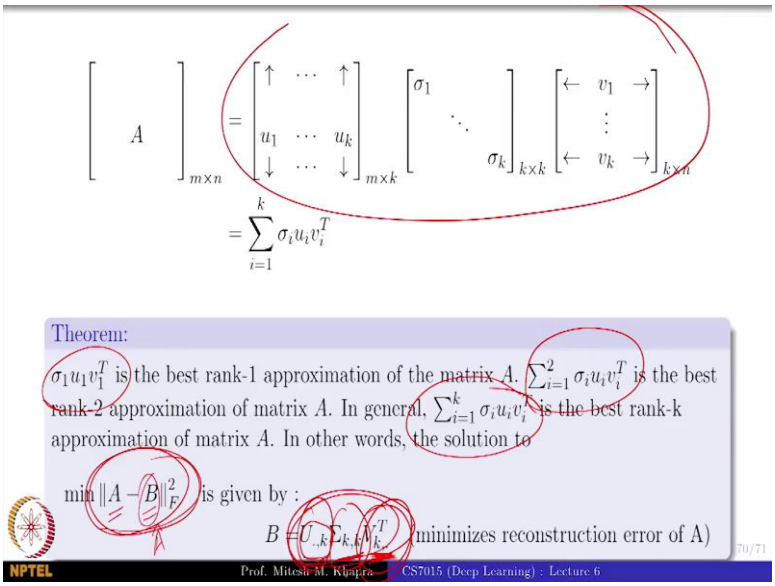
K so, if k is very less than your m and n, right? Then again you will have some compression you get this ok. So, all of these ideas are related and I want you to be able to connect them, right? That all of this is towards doing some approximations reconstructing some reconstructing a matrix from it is components, and doing this reconstruction in a manner that you end up making minimum error in the reconstruction. Is this idea clear? Even if some part of the math is not clear, is this idea clear? How many forget this? Ok so, some of you do not, you do not?

Student: (Refer Time: 20:59).

Yeah so, what is the original dimension of A? M cross n, right now I am trying to reconstruct it using a sum of sum k terms, ok. So, hence this k comes, now each of these terms how many elements do I have? I have ui which is of dimension m, I have v i which is of dimension n and then I have the sigma i which is of dimension one, right? And I have k of these. So, this is the total amount of storage that I need. I am saying that as k is much less than m and n which would typically be the case.

Then you are getting a much lower space reconstruction of the original data, right? And you are doing this reconstruction smartly, because you are not taking any k dimensions, you are taking the k most important dimensions, and this most important is defined by the singular values, this is designed by the sigma is that fine?

(Refer Slide Time: 22:02)



Ok and actually there is a formal theorem which says that sigma 1 u 1 v 1 transpose is the best ranked one approximation of the matrix, is this a rank one matrix? is sigma 1 u n I hope you guys have done the assignment, right? Sigma 1 u 1 v 1 transpose is the rank one matrix. And if I take this idea further, this summation is the best ranked 2 approximation and if I keep going, this summation is the best rank k approximation. So, what it says is that if you are trying to reconstruct the original matrix, right from these components. And if you go by the Eigen or the singular values, and you pick the ones corresponding the top k singular values then the best that is the best possible reconstruction that you could have done.

Now, how do you formally define reconstruction? How would you make it as an optimization problem? What are you trying to minimize?

Student: (Refer Time: 22:59).

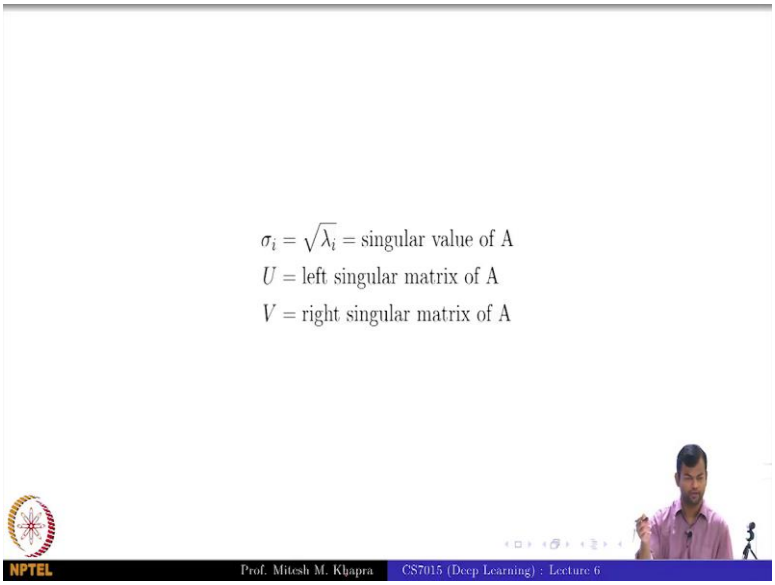
The actual matrix has some values which is the matrix A, ok. B is the reconstructed matrix using only k dimensions, how many of you understand? What is this product saying what is this?

Student: (Refer Time: 23:20).

First k columns of u. These are the k singular values, and these are the first k rows of. So, ok I was just talking about this is the first k columns of u, these are the k singular values

are put across the diagonal, and this is the first k rows of V transpose. And this is exactly the product which I showed you here, is that fine?

(Refer Slide Time: 23:49)



Ok so, there is a theorem this is called the SVD theorem it says that, if you want to reconstruct a then this is the best rank k approximation that you can get. Now if I want to pose it as an optimization problem what will I say what would I have minimized actually. This is the reconstruction, right? So, let us call it a hat actually, and what does this mean? This is the dash norm.

Student: (Refer Time: 24:15)

Frobenium norm, what does the Frobenium norm give you? Squared difference between the elements, right? Roughly speaking ok; so, it will tell you what is the square difference between the ij'th element of a and the ij'th element of B. So, whenever we have this situation if you are trying to if this is our objective function that we have trying to reconstruct A, or try to transform something and get a predicted A or A reconstructed A then the best possible reconstruction would be given by this solution.

So, this optimization problem has a solution that you just use the eigenvectors of xx transpose and sorry AA transpose and A transpose A, right? Is that clear? Ok so, this is the theorem that we will be using when we are talking about autoencoders, and we will try to connect auto encoders to PCA ok. So, just revise this is the prerequisite for next

class whatever we have done in the last 3 sort of extra lectures, you have to revise it before you come for class tomorrow, right, ok. And yeah this is sigma is just some terminology sigma is actually the square root of lambda A that was obvious, and U is called the left singular matrix of a and V is called the right singular matrix of A.